



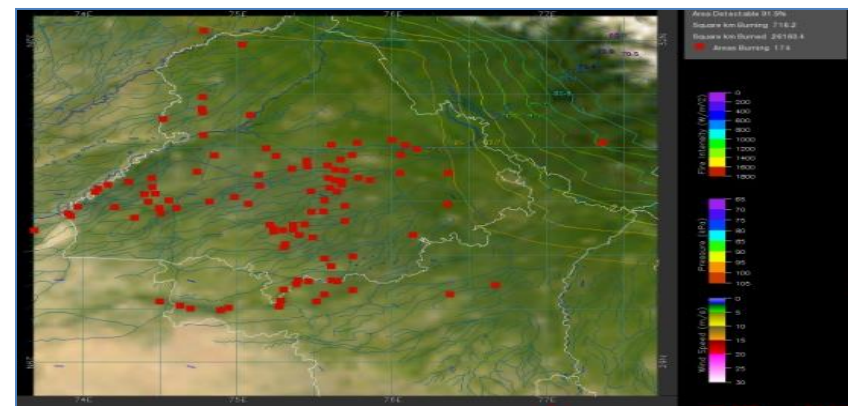
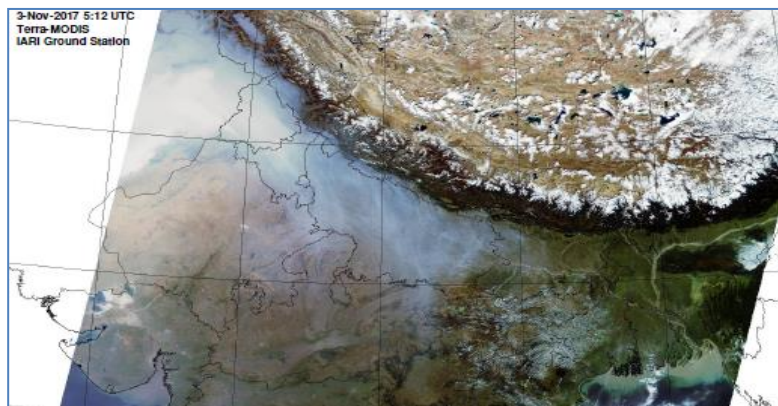
Bulletin No. 26

Events Date:
10-Oct-2023

Issued on:
10-Oct-2023



MONITORING PADDY RESIDUE BURNING IN INDIA USING SATELLITE REMOTE SENSING DURING 2023



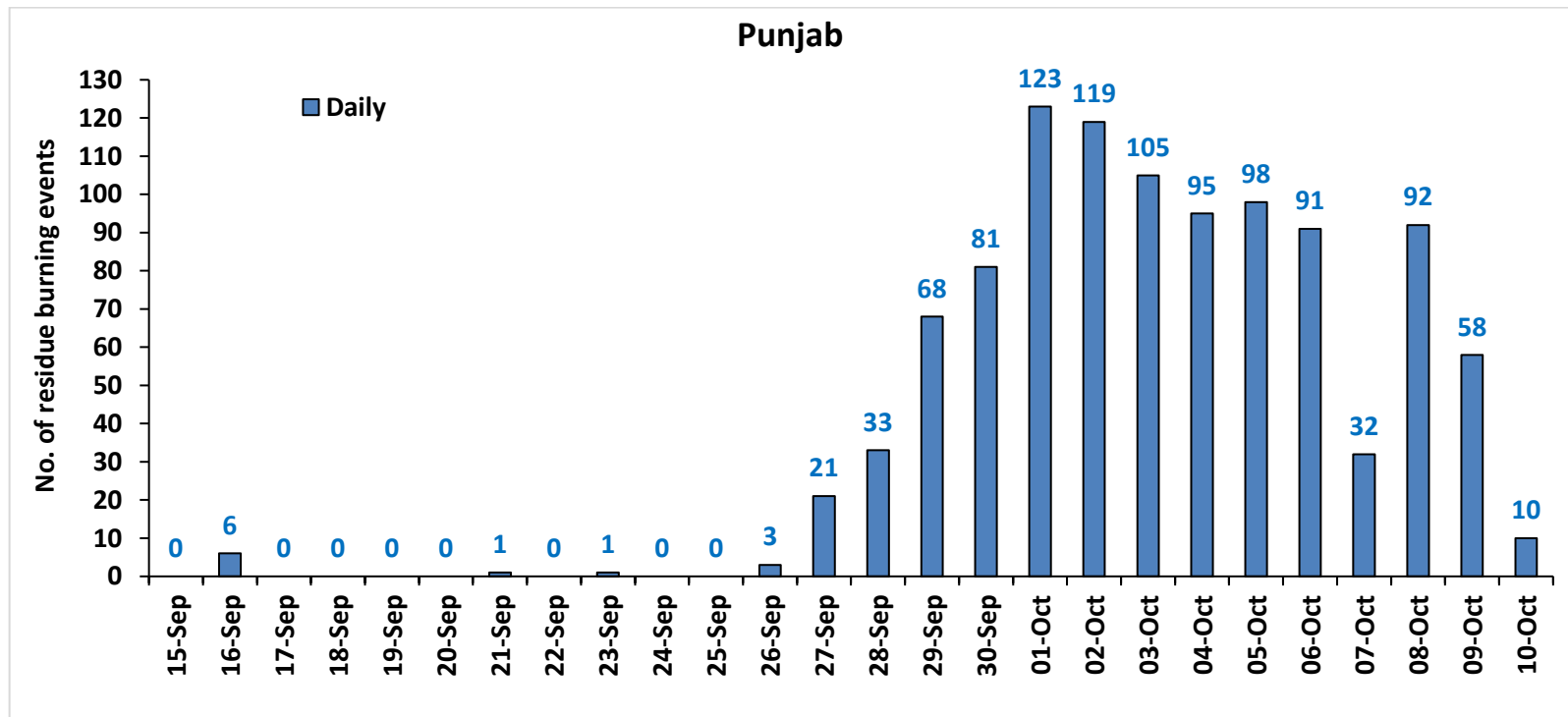
Consortium for Research on Agroecosystem Monitoring and Modeling from Space (CREAMS) Laboratory,
Division of Agricultural Physics, ICAR – Indian Agricultural Research Institute, New Delhi – 110012

<http://creams.iari.res.in>

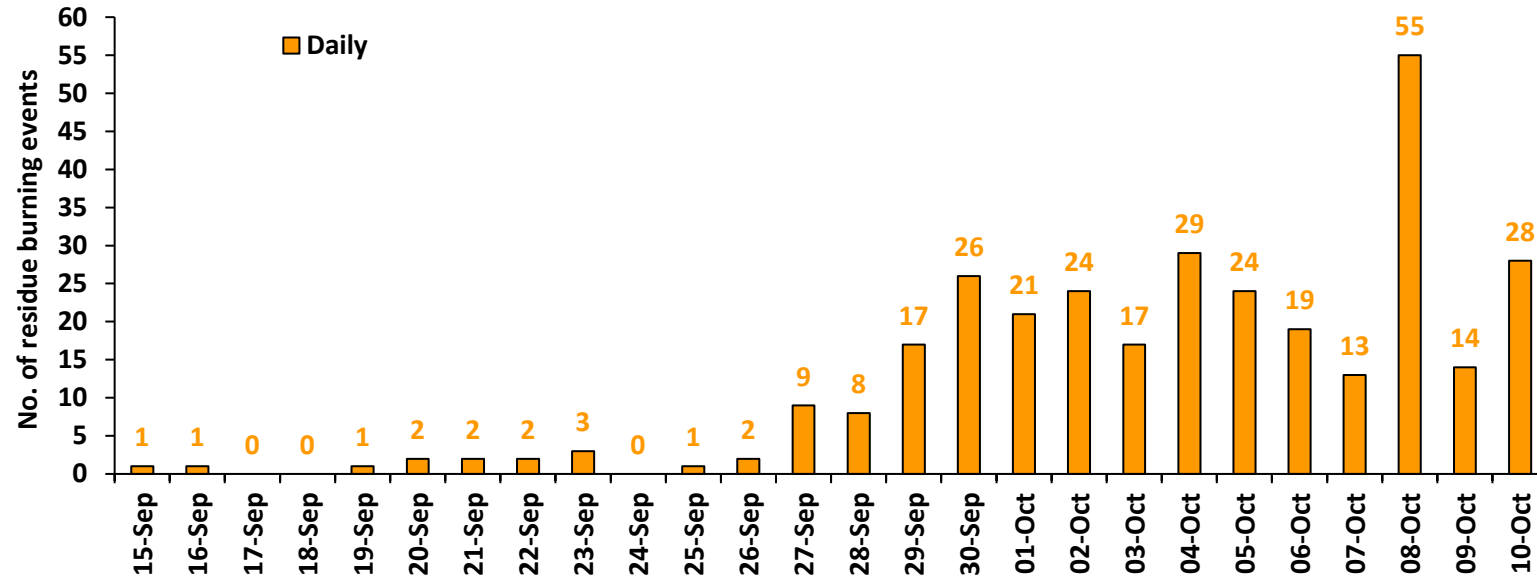
Highlights for 10-Oct-2023

- The active fire events due to rice residue burning were monitored using satellite remote sensing, following the "Standard Protocol for Estimation of Crop Residue Burning Fire Events using Satellite Data".
- Satellites detected **150** residue burning events in the six study States on 10-Oct-2023.
- The state wise events detected on 10-Oct-2023 were **10** in Punjab, **28** in Haryana, **34** in UP, **00** in Delhi, **26** in Rajasthan and **52** in MP.
- Total **1841** burning events were detected in the six states between 15-Sept-2023 and 10-Oct-2023, which are distributed as **1037**, **319**, **185**, **01**, **124** and **175** in Punjab, Haryana, UP, Delhi, Rajasthan and MP, respectively.

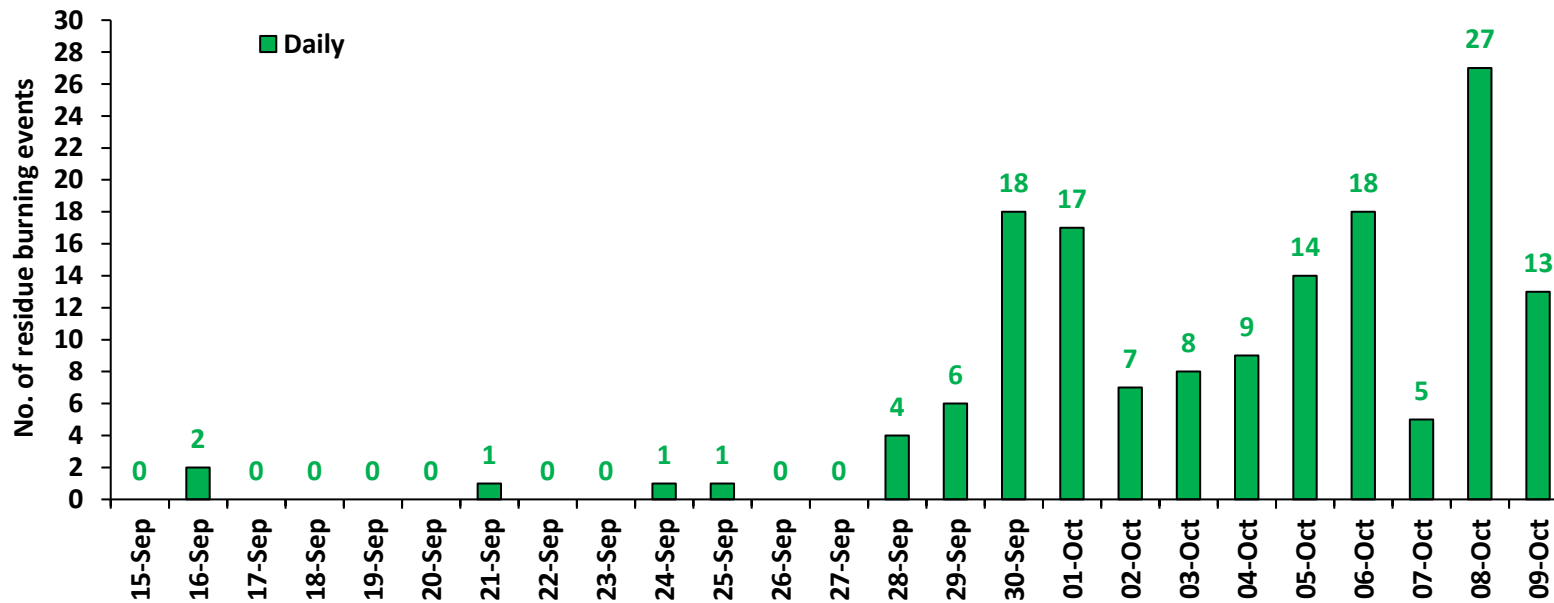
Temporal distribution of rice residue burning events for the study States in 2023

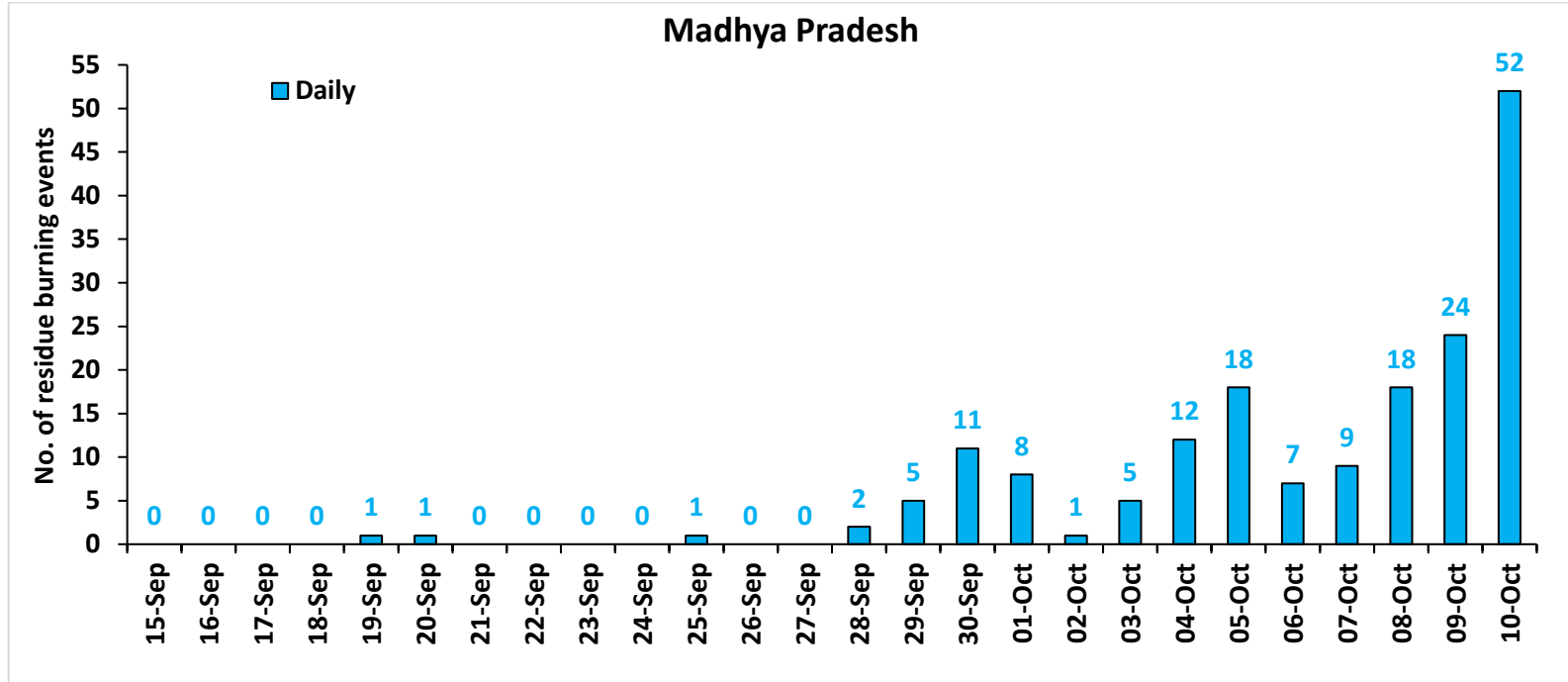
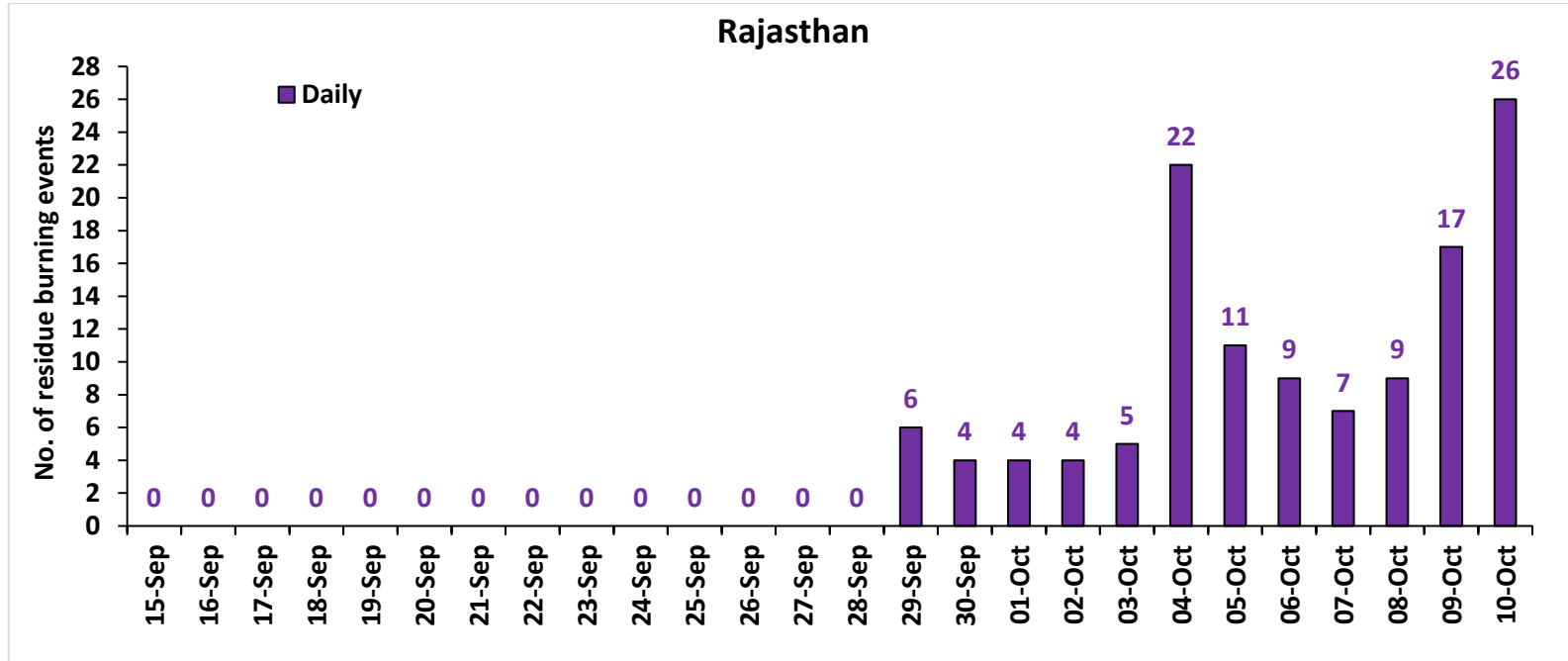


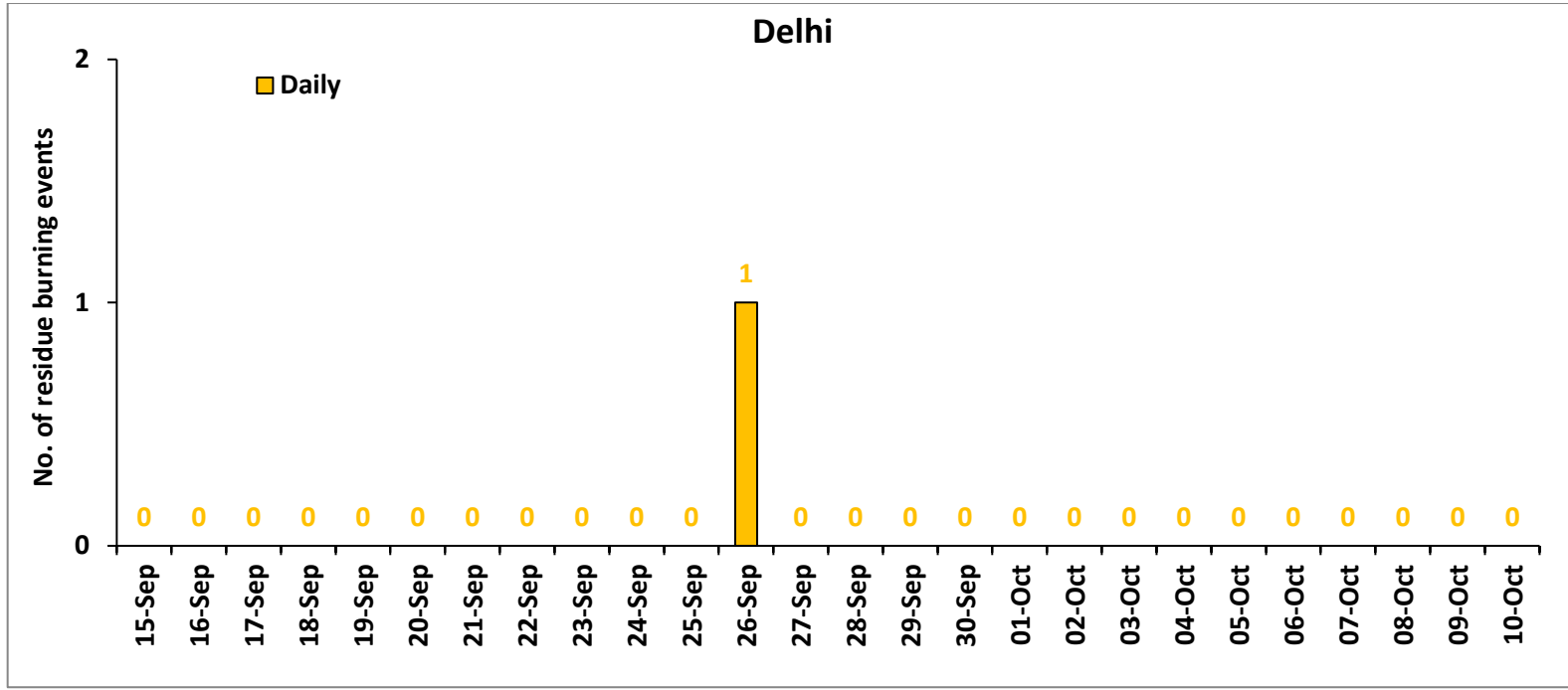
Haryana



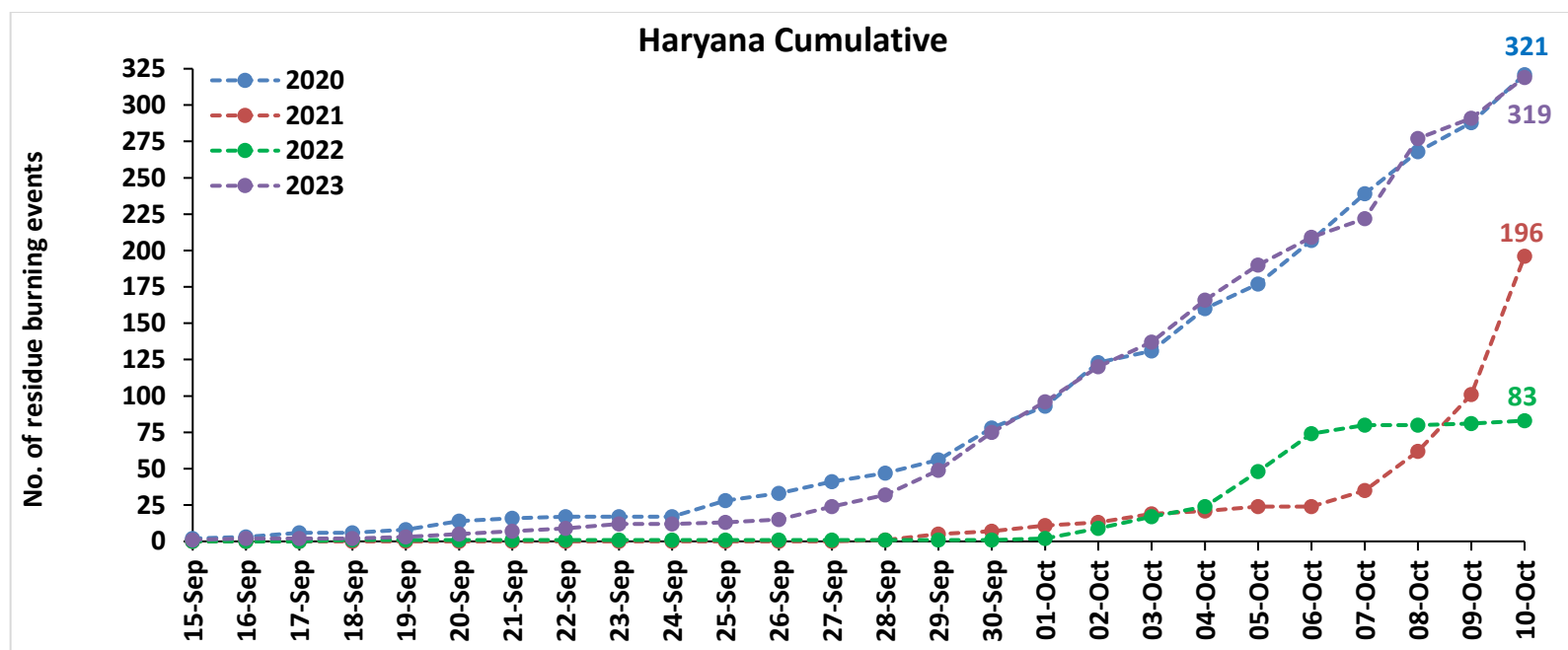
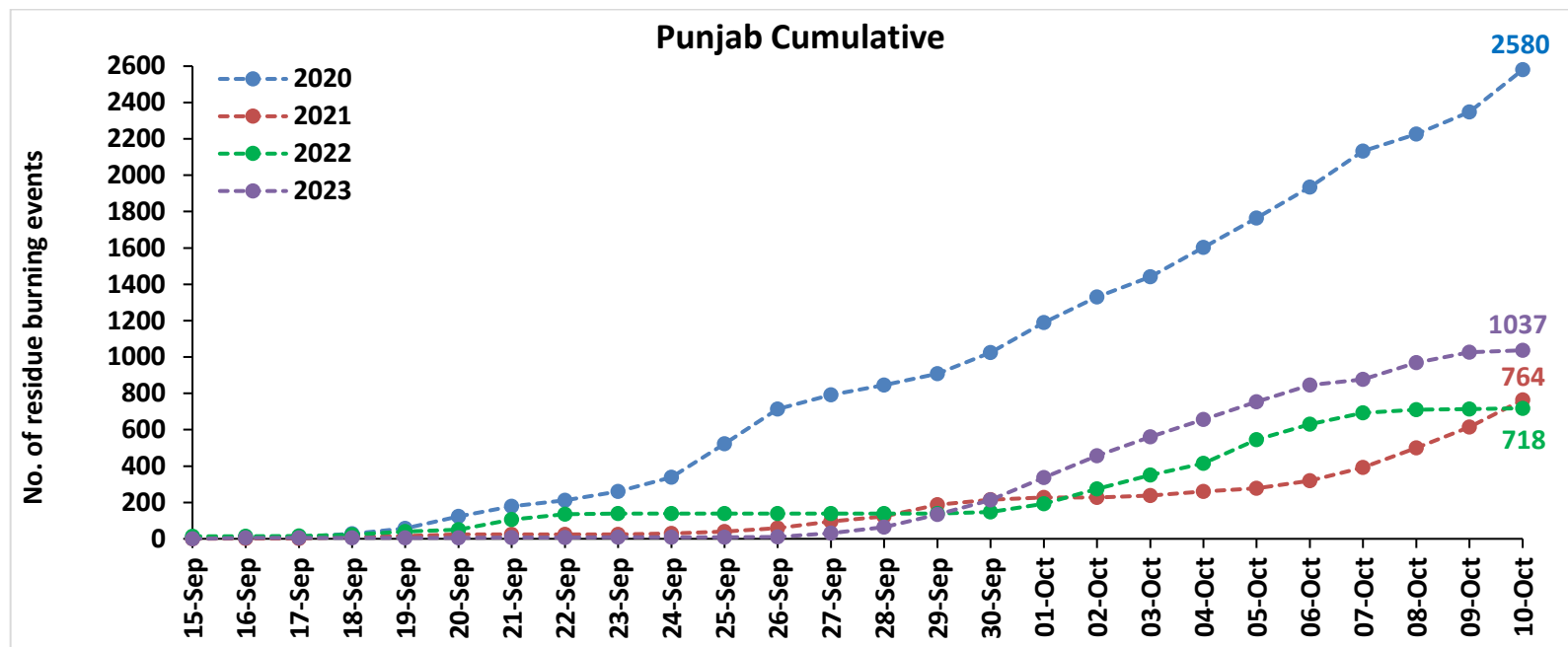
Uttar Pradesh

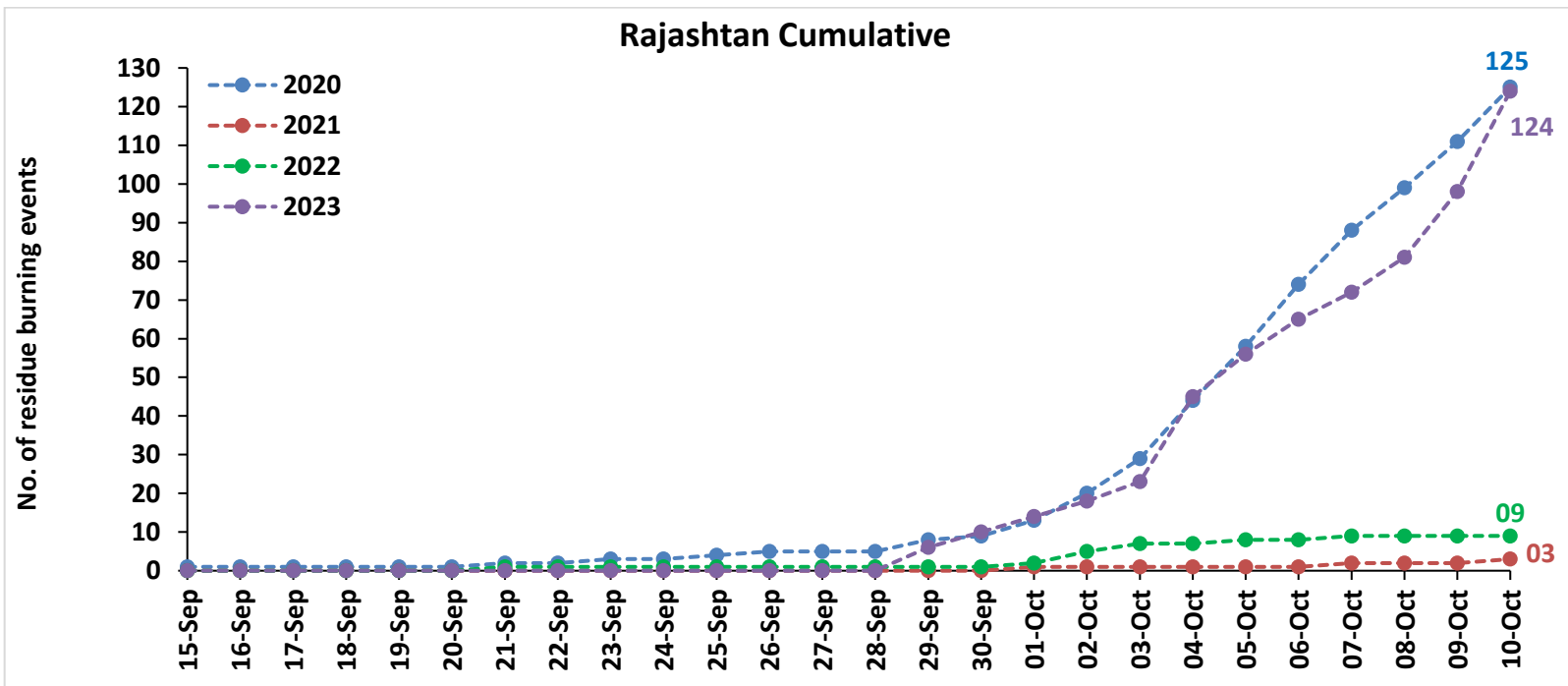
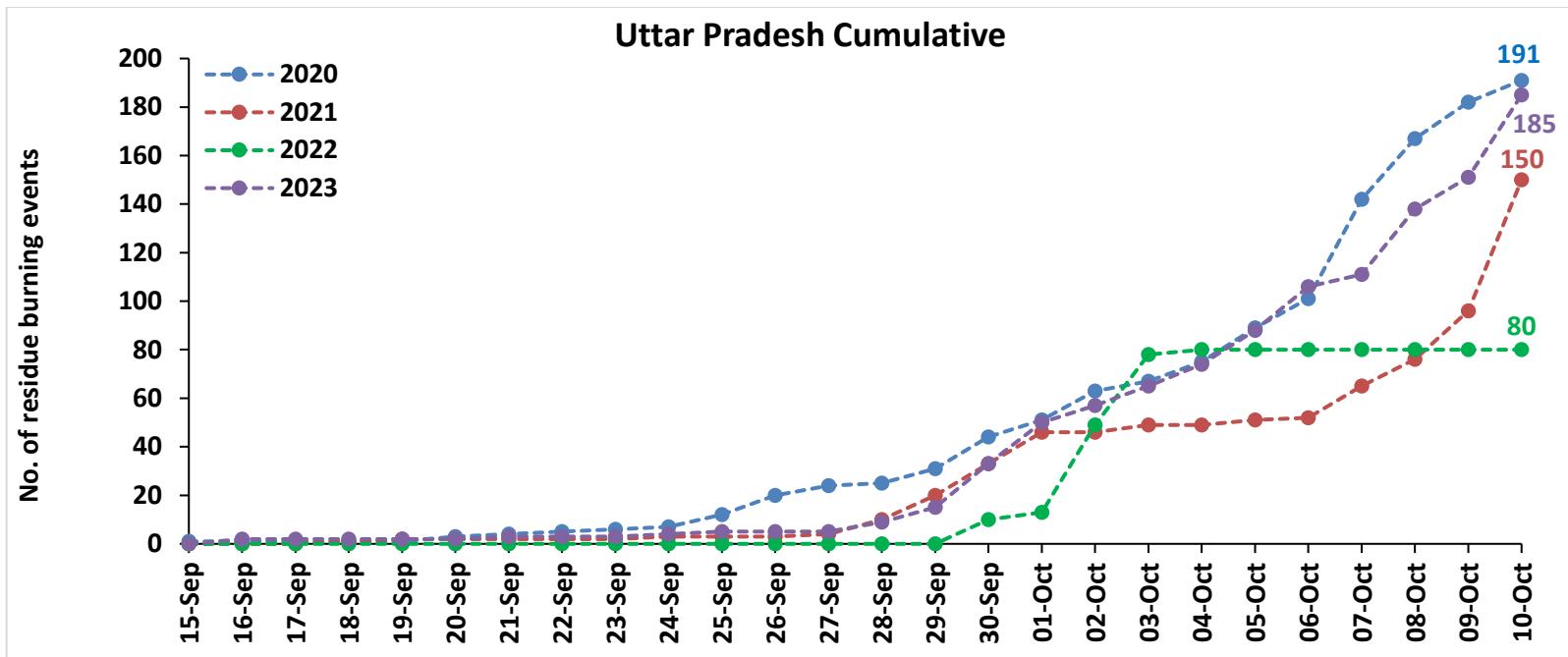


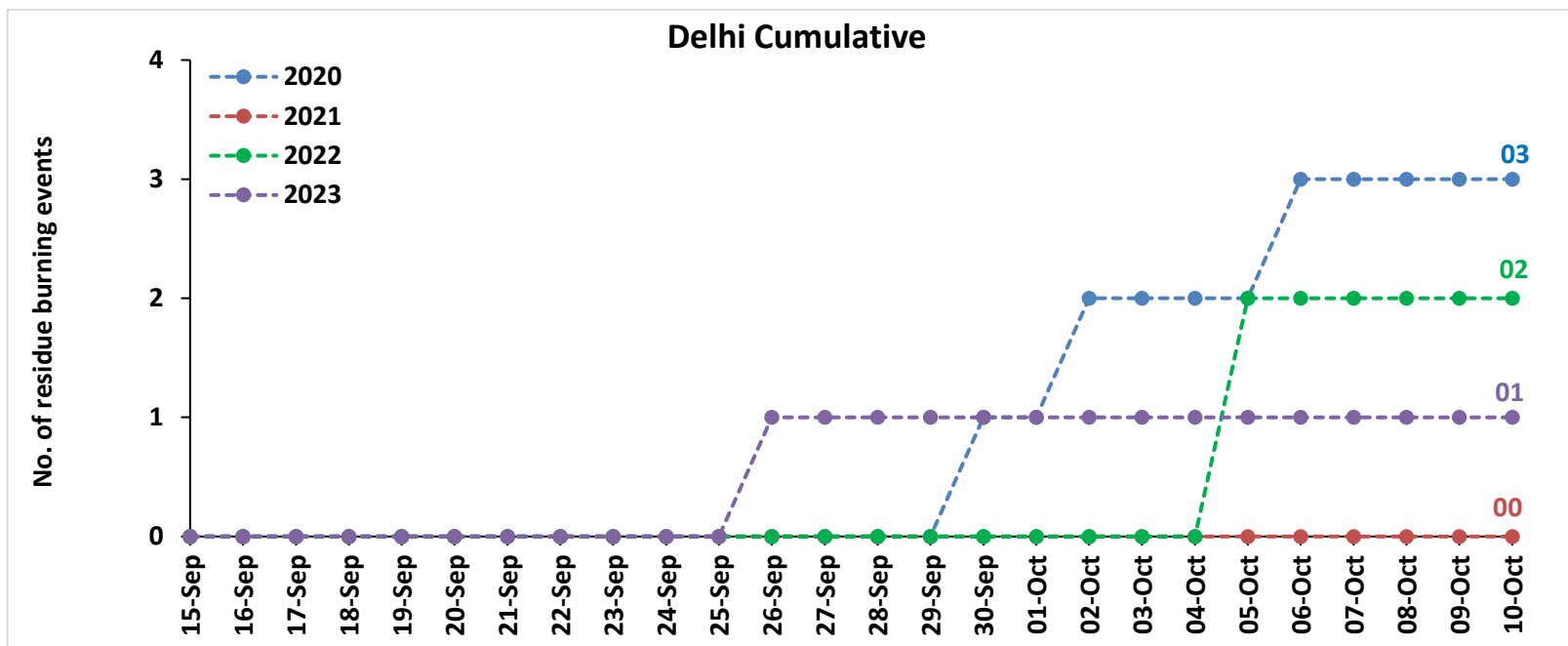
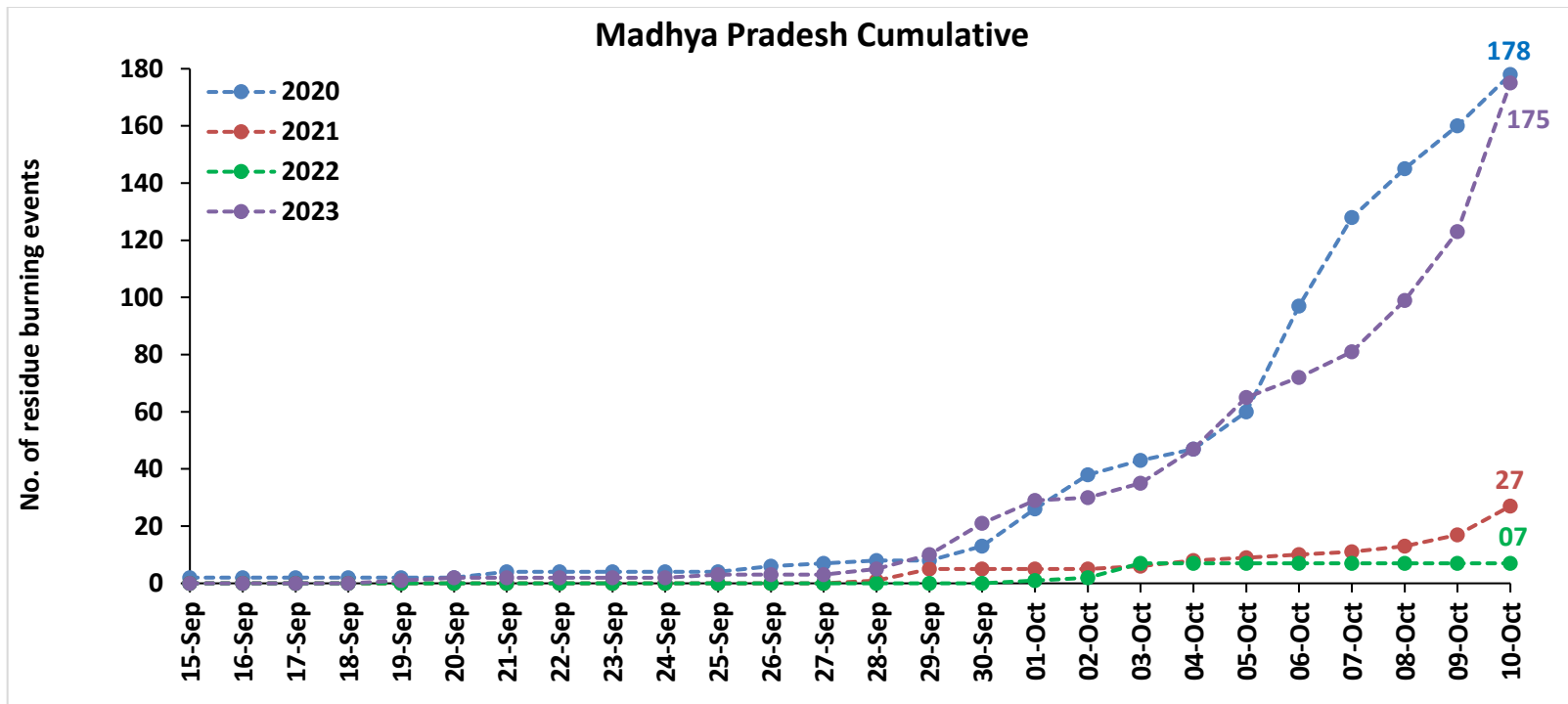




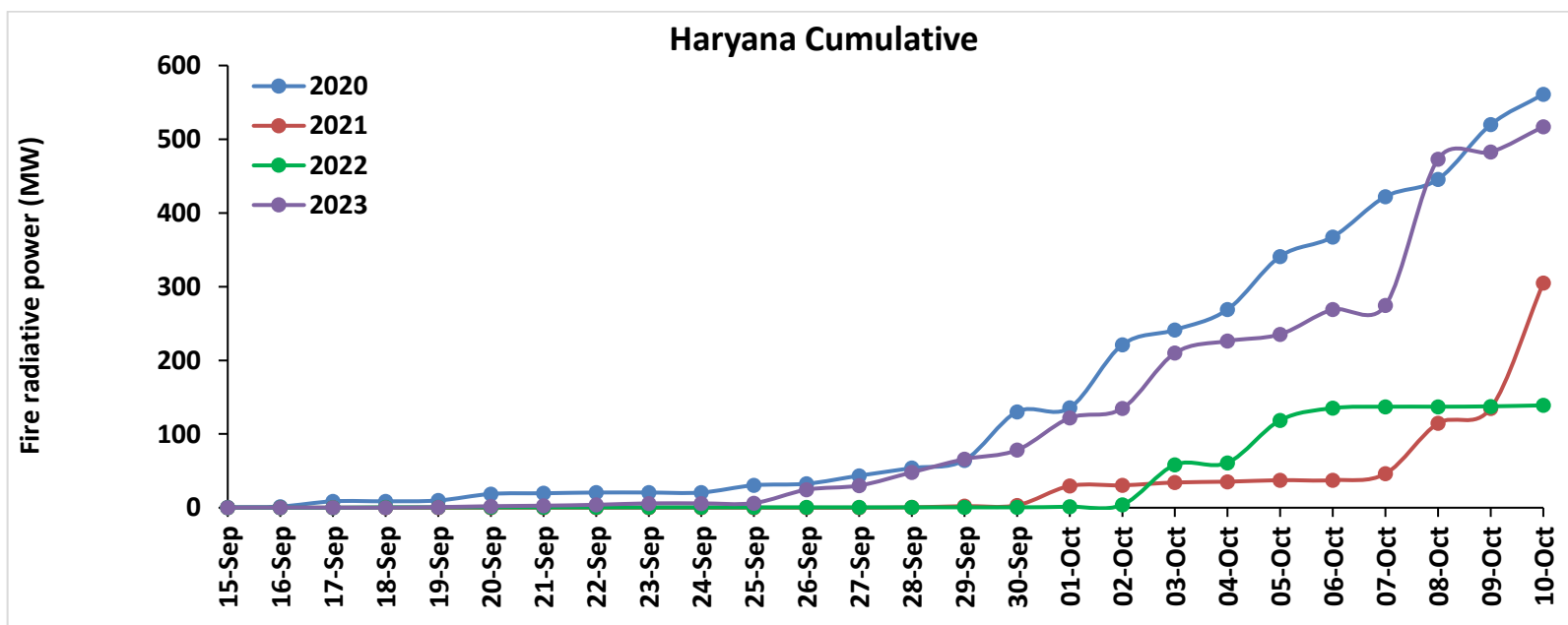
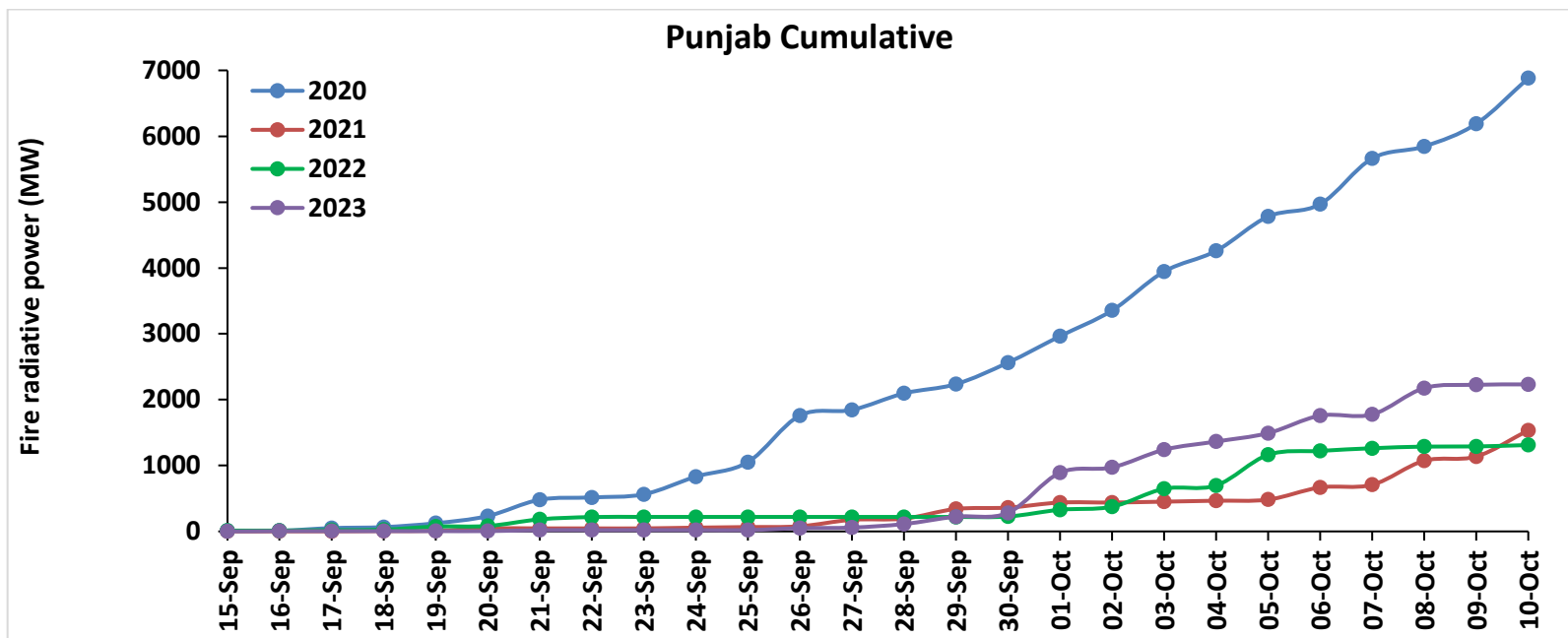
Comparison of residue burning events in current year (2023) with previous years (2020, 2021 and 2022) for the study State

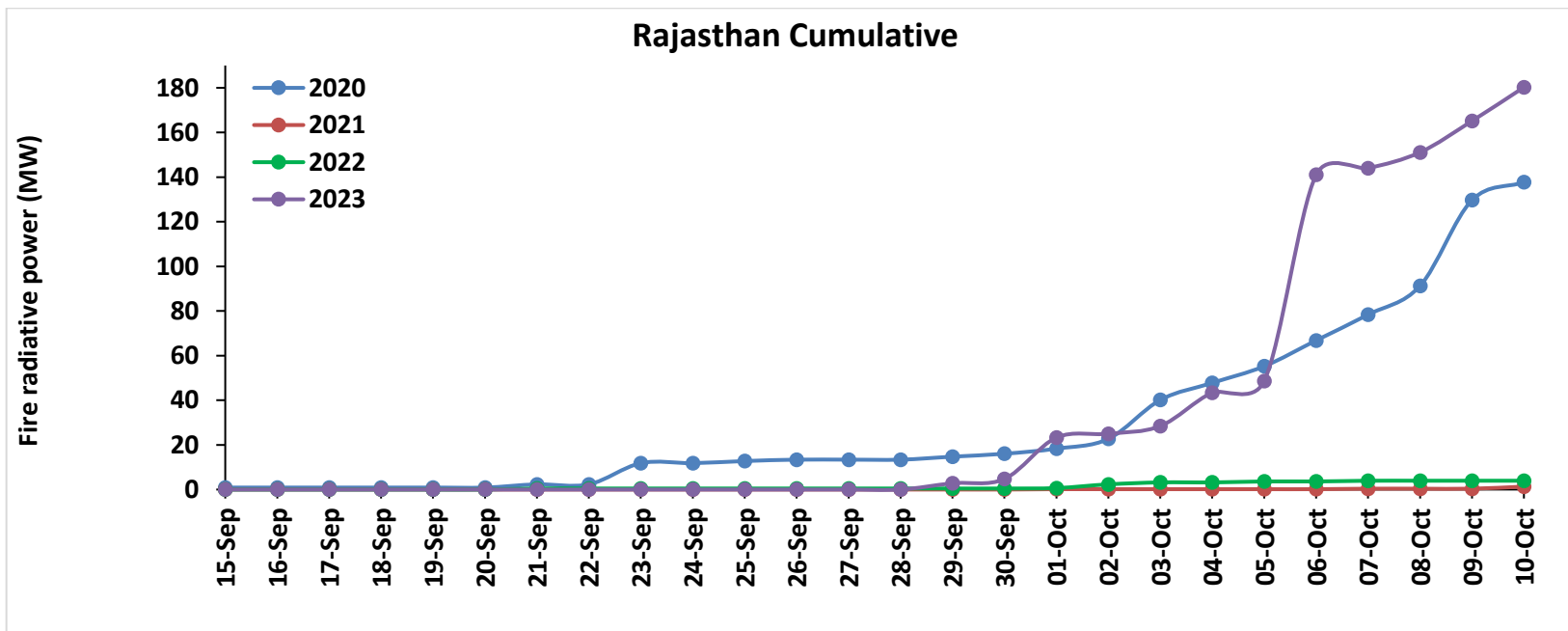
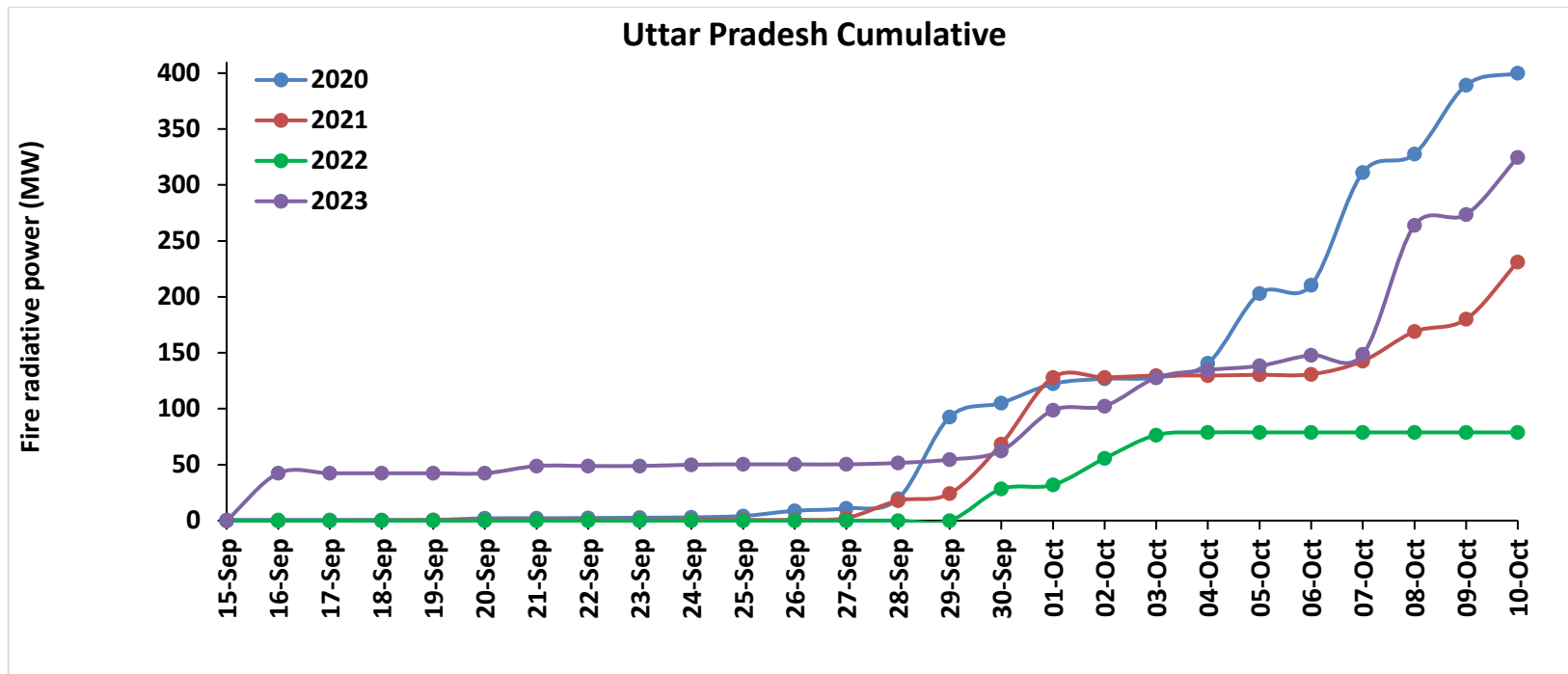


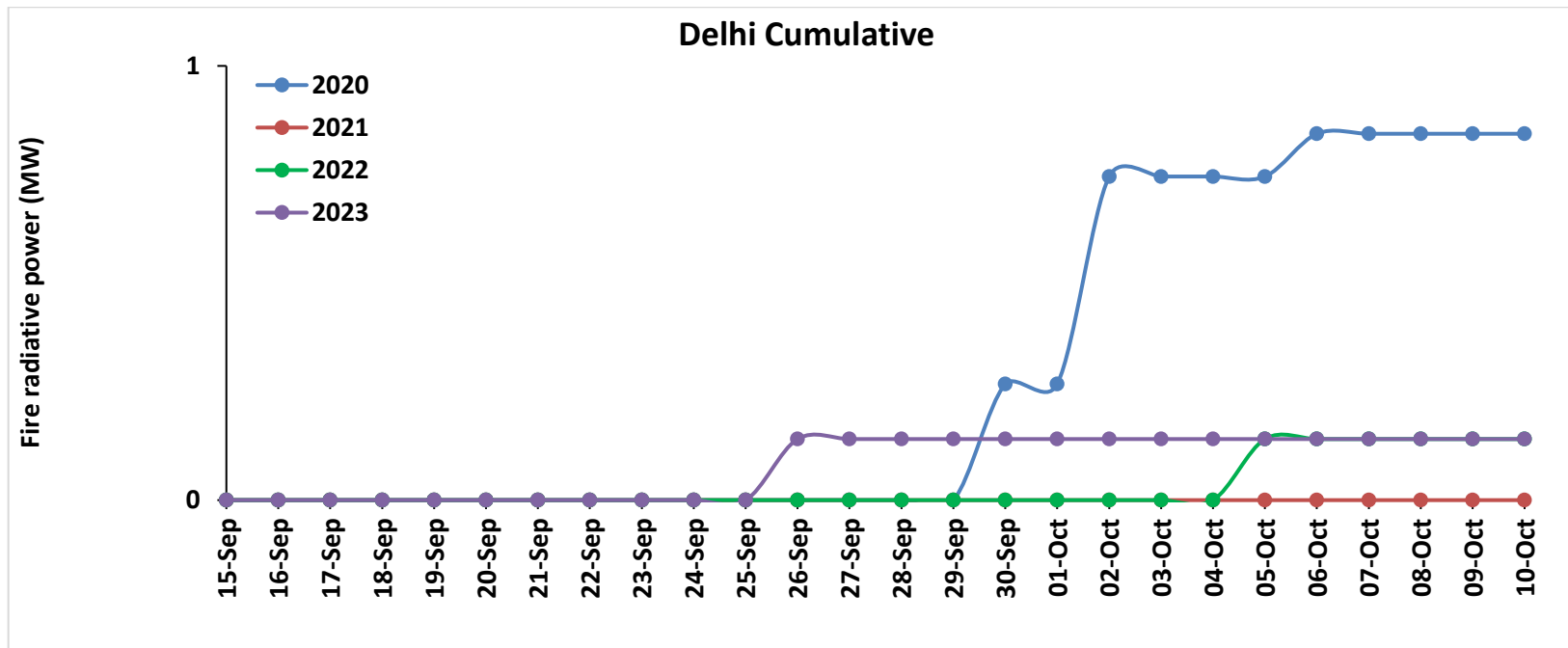
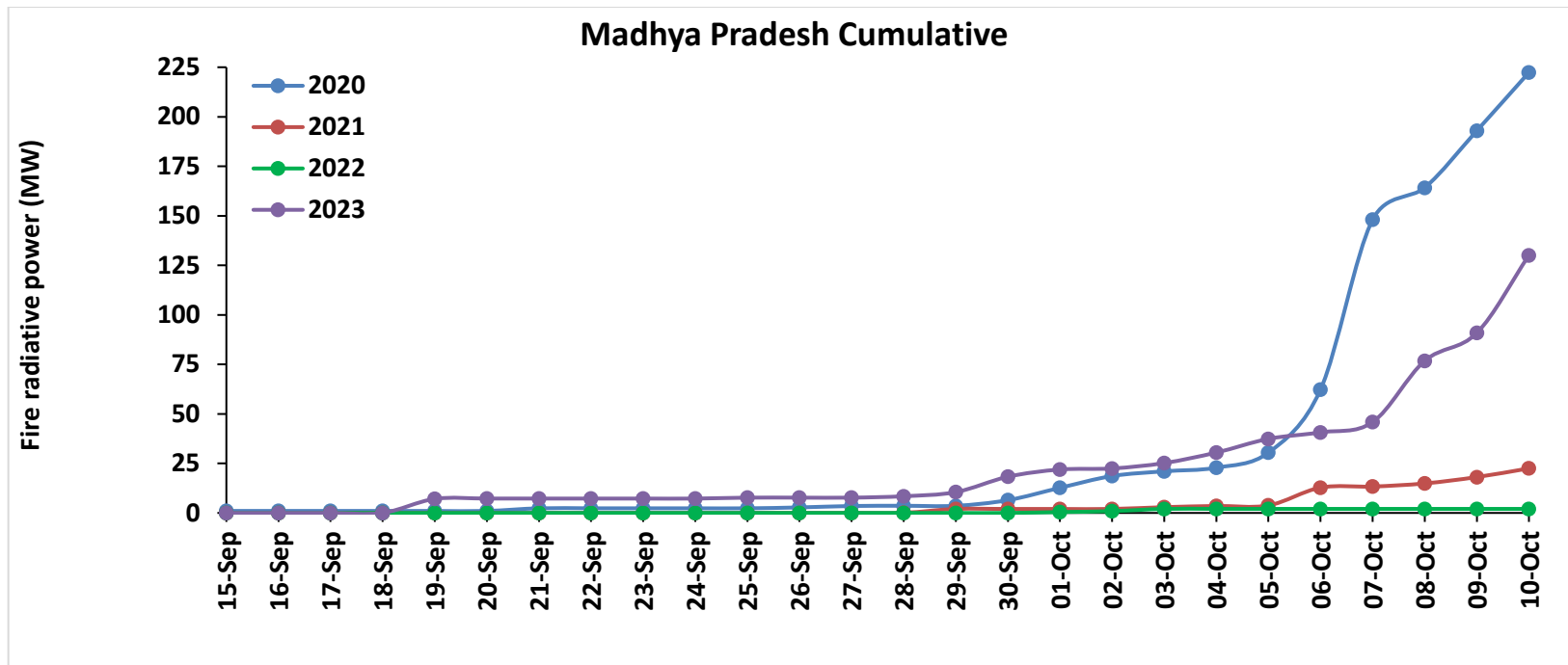




Comparison of residue burning fire power in current year (2023) with previous years (2020, 2021 and 2022) for the study States

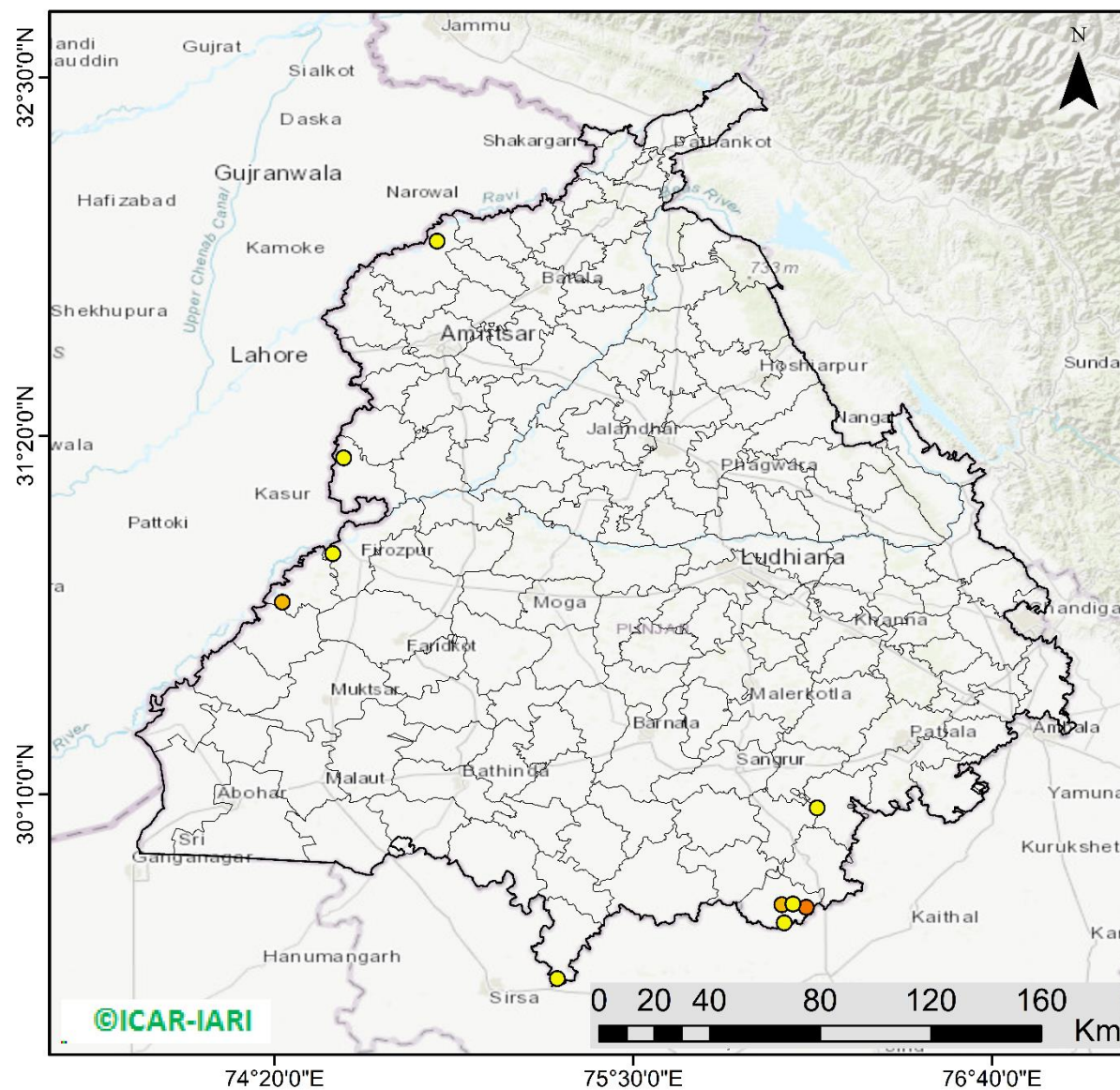






Punjab

RICE RESIDUE BURNING IN PUNJAB



10 burning events
detected in Punjab during
(10th Oct 2023)

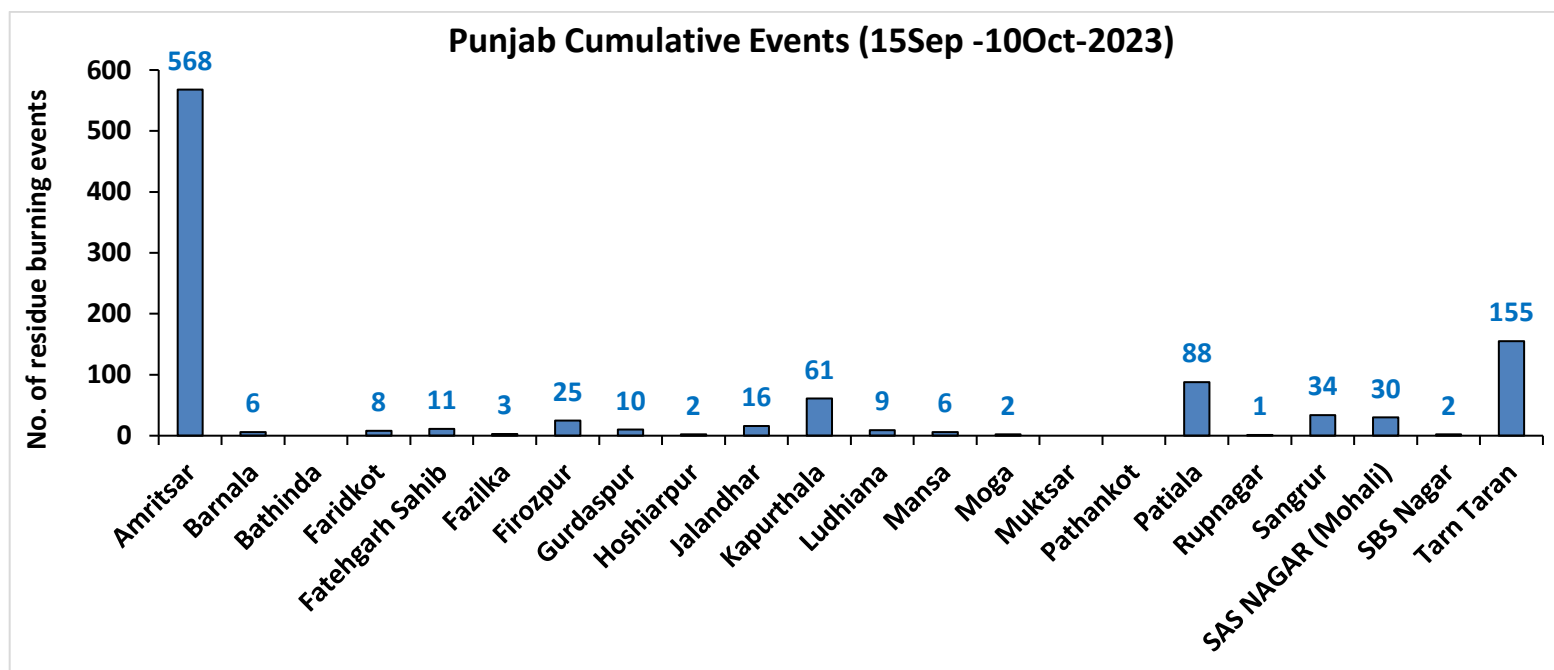
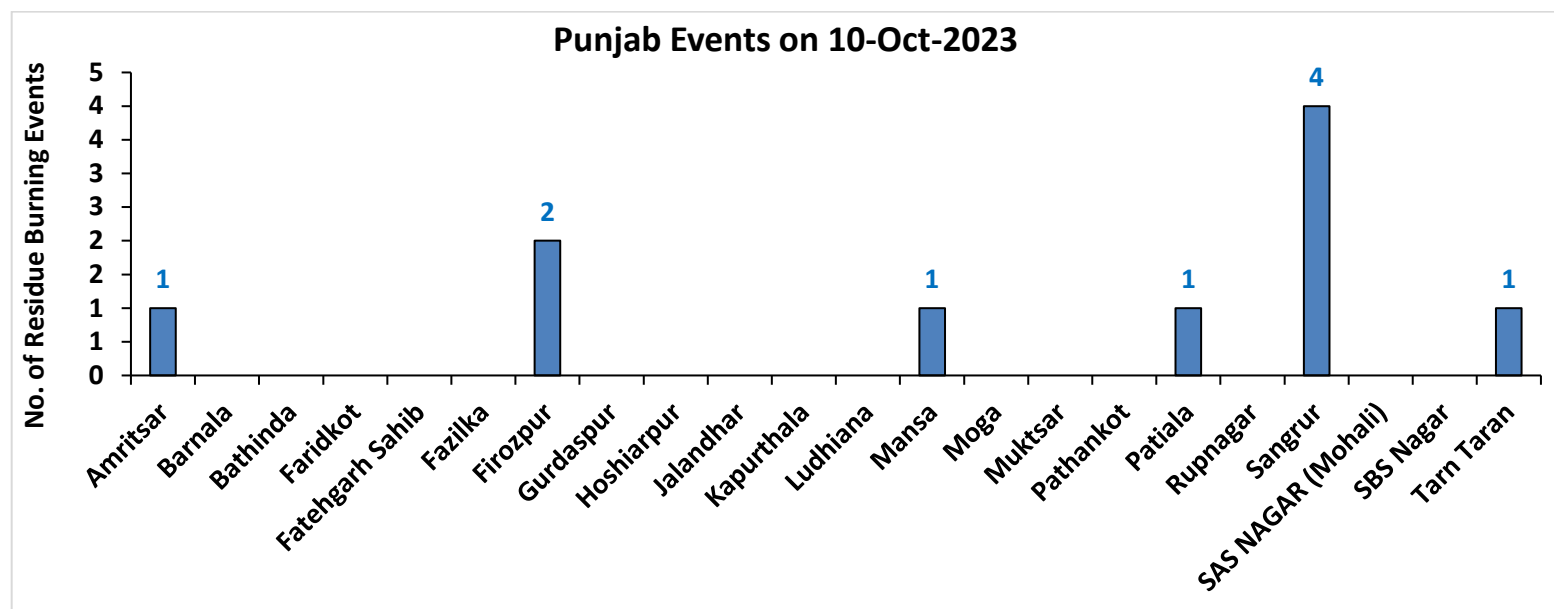
Fire Intensity (W/m²)

- 0 - 5
- 6 - 10
- 11 - 15
- 16 - 20
- >20



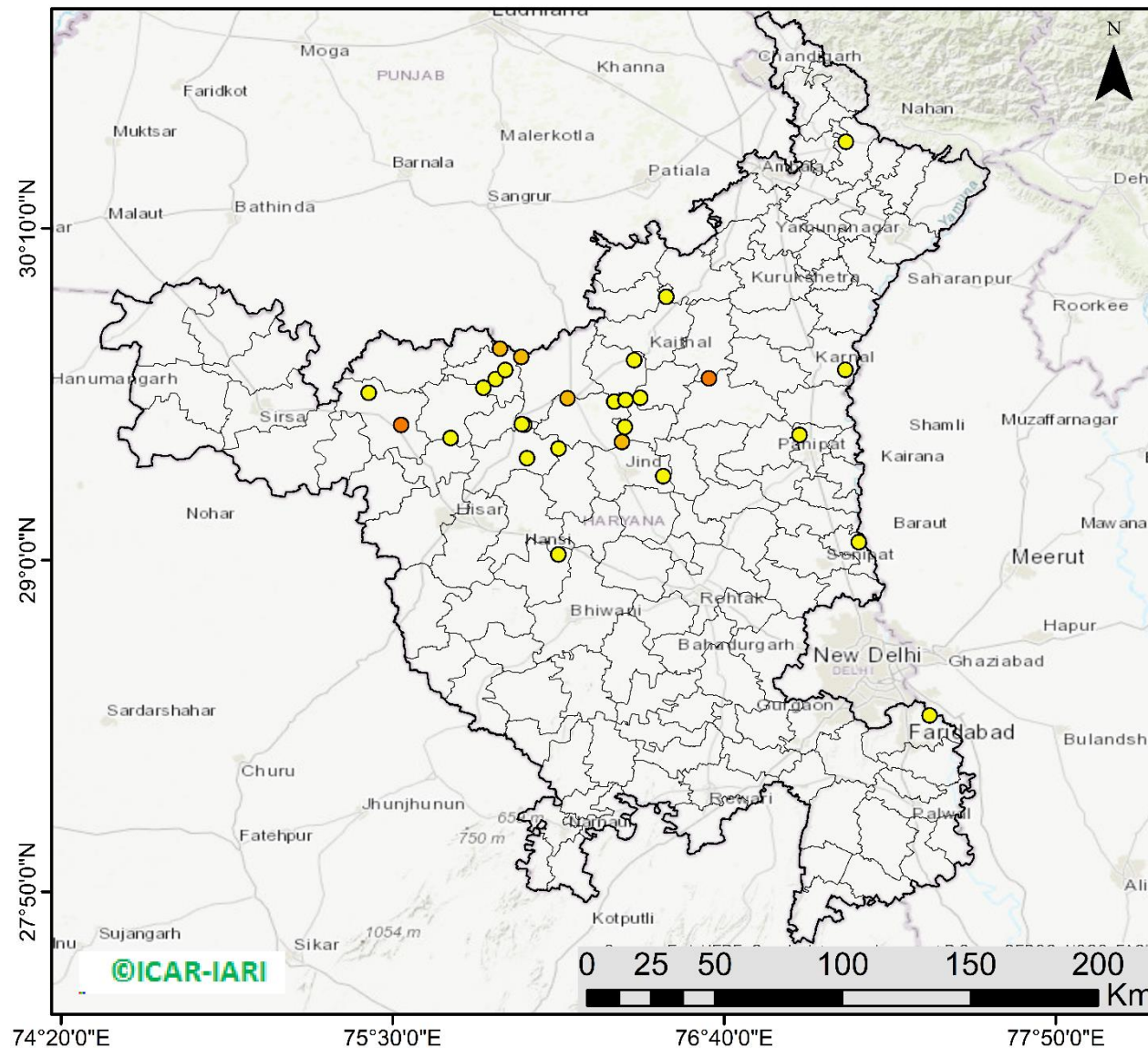
<http://geoportal.icar.gov.in:8080/geoexplorer/composer/>

District-wise cumulative number of residues burning events in Punjab (15Sep- 10Oct-2023)



Haryana

RICE RESIDUE BURNING IN HARYANA



28 burning events
detected in Haryana on
(10th Oct 2023)

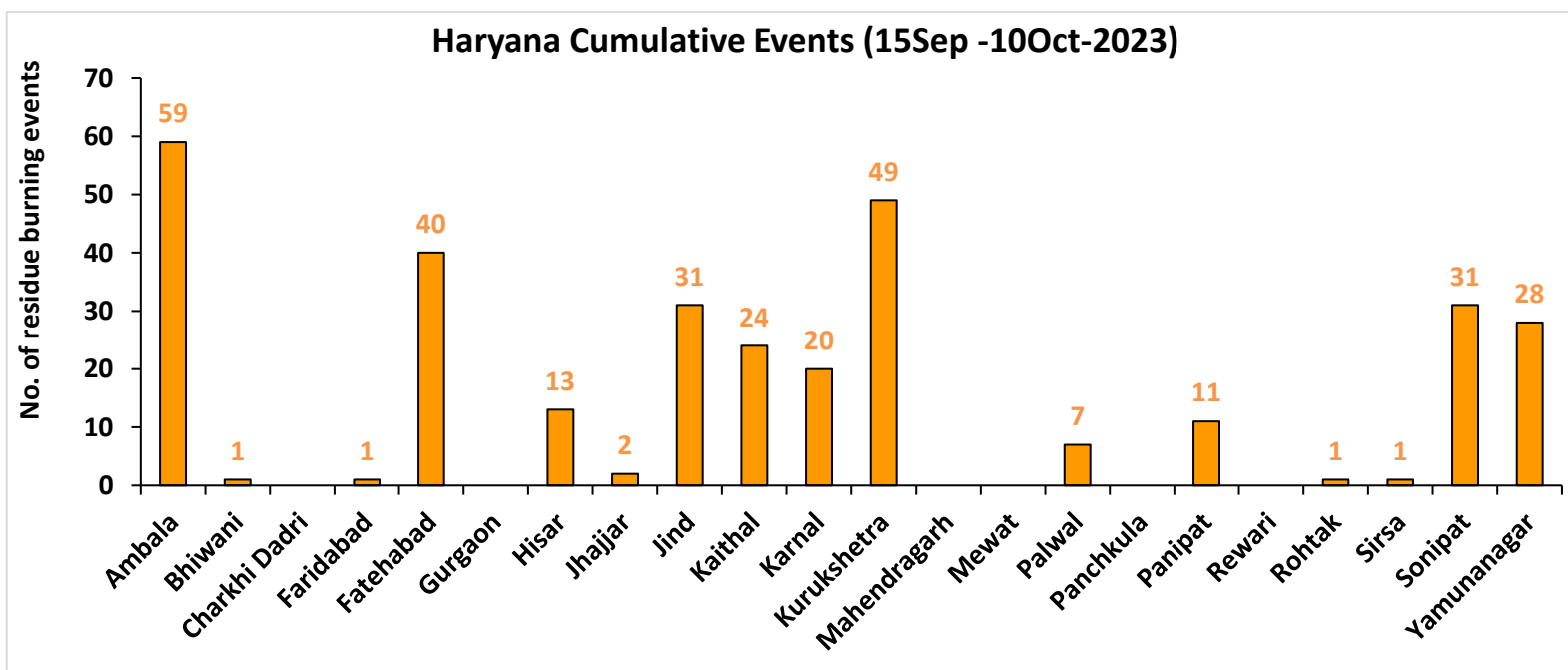
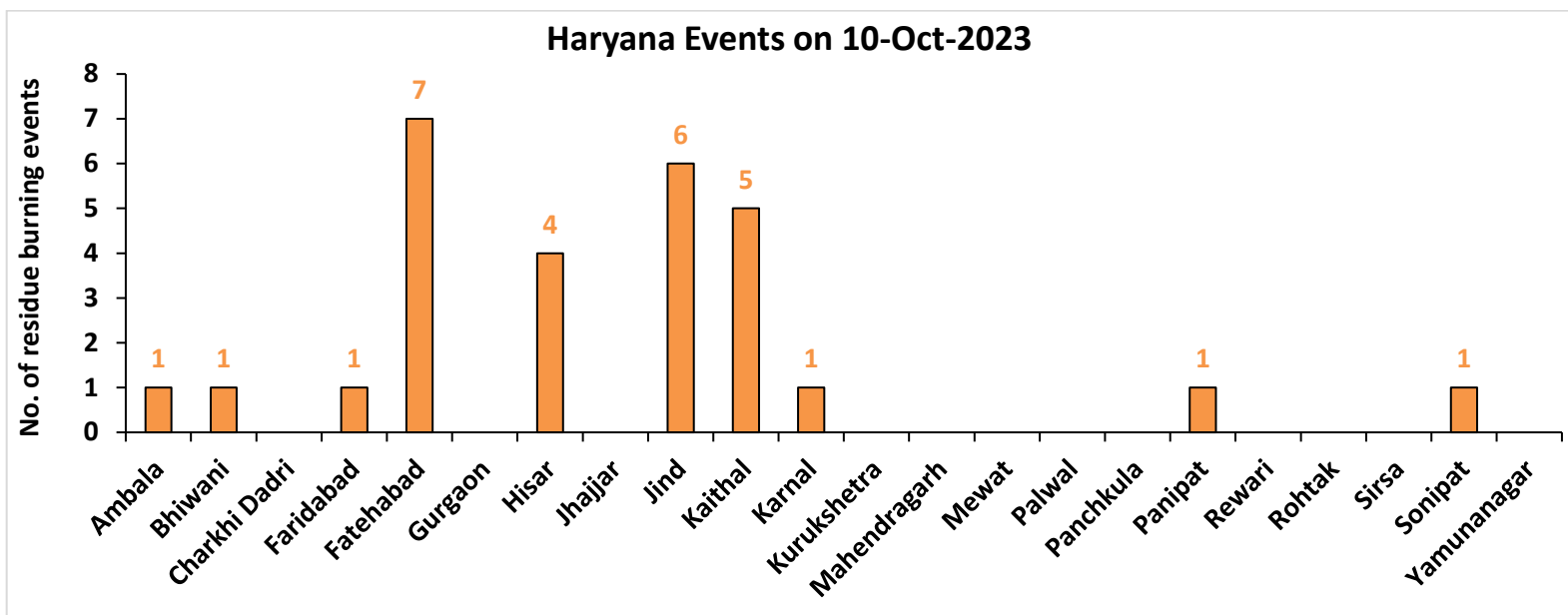
Fire Intensity (W/m²)

- 0 - 5
- 6 - 10
- 11 - 15
- 16 - 20
- >20



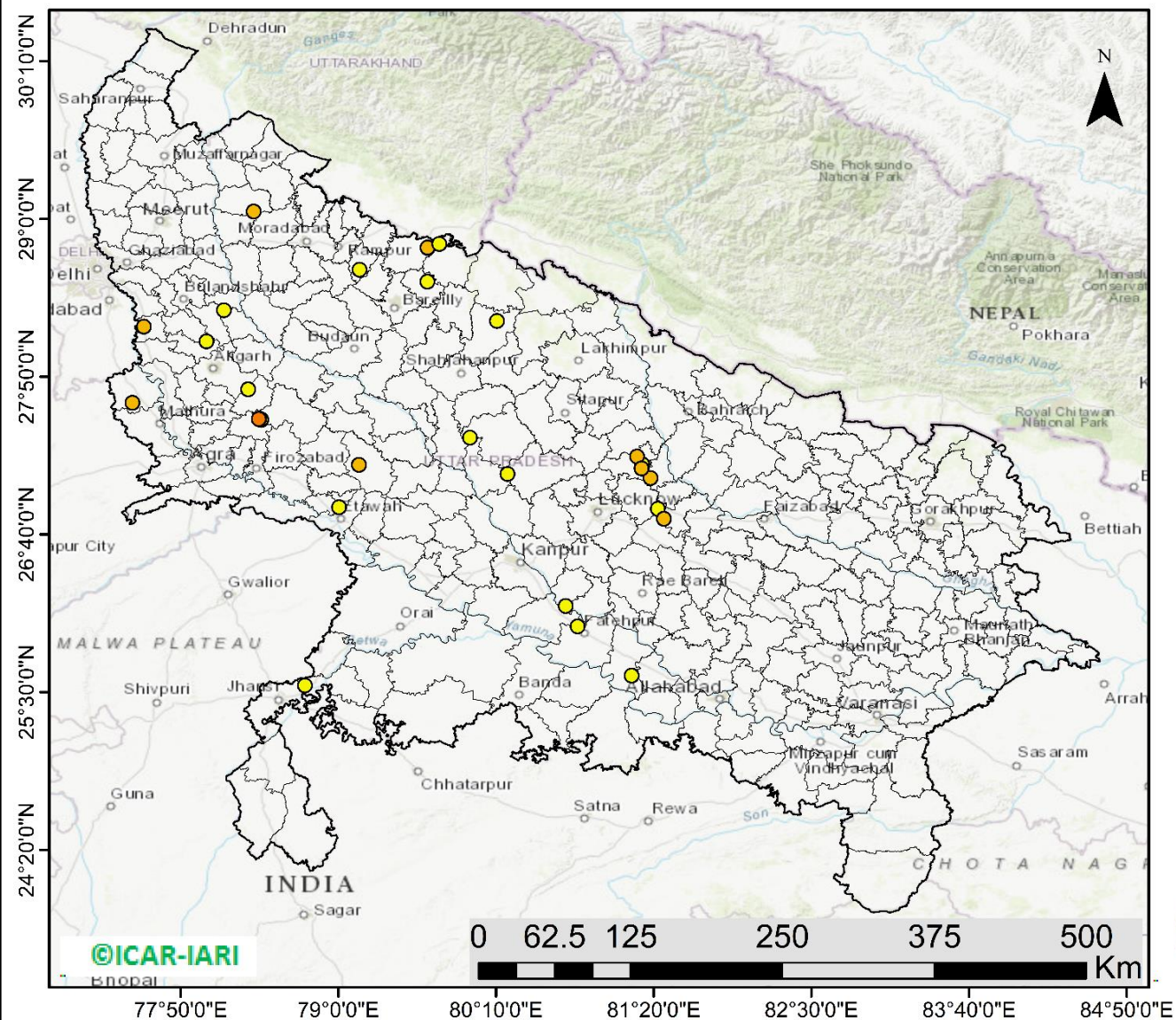
<http://geoportal.icar.gov.in:8080/geoexplorer/composer/>

District-wise cumulative number of residues burning events in Haryana (15Sep-10Oct -2023)



Uttar Pradesh

RICE RESIDUE BURNING IN UTTAR PRADESH



34 burning events
detected in Uttar Pradesh on
(10th Oct 2023)

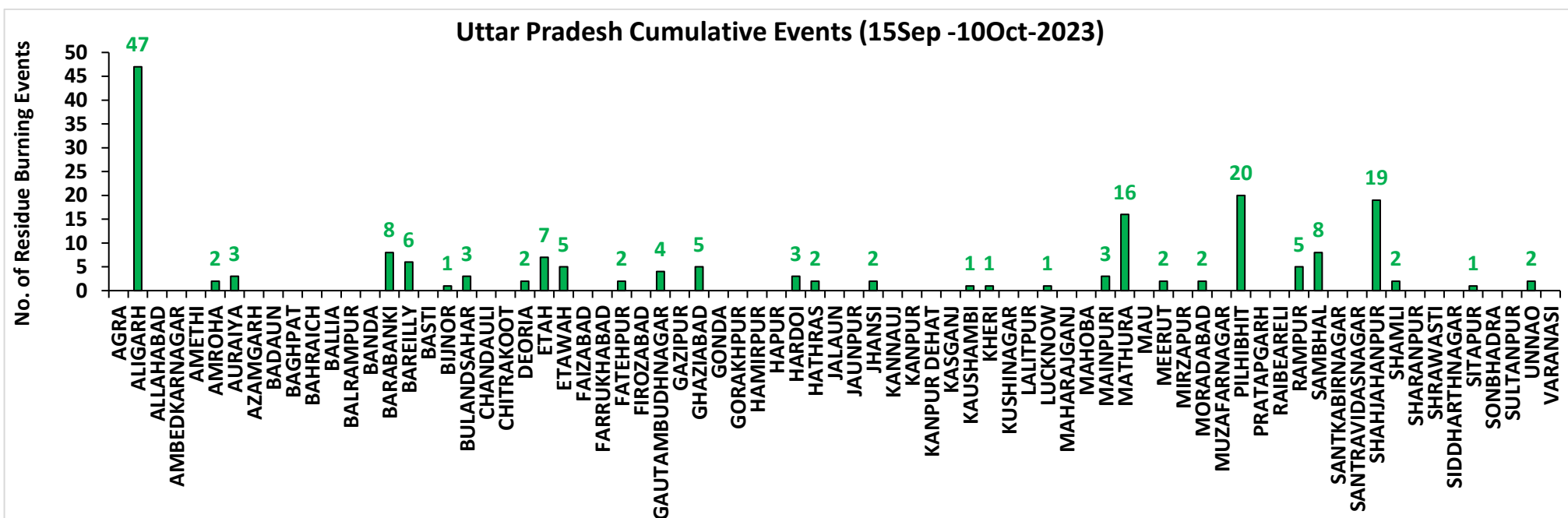
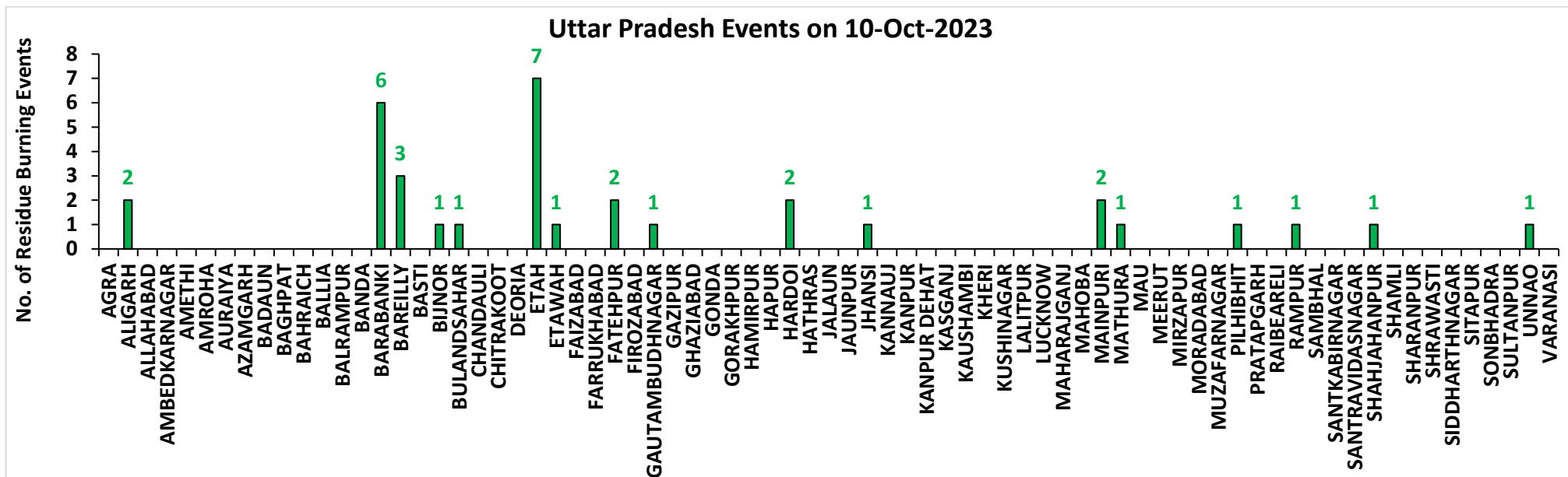
Fire Intensity (W/m²)

- 0 - 5
- 6 - 10
- 11 - 15
- 16 - 20
- >20



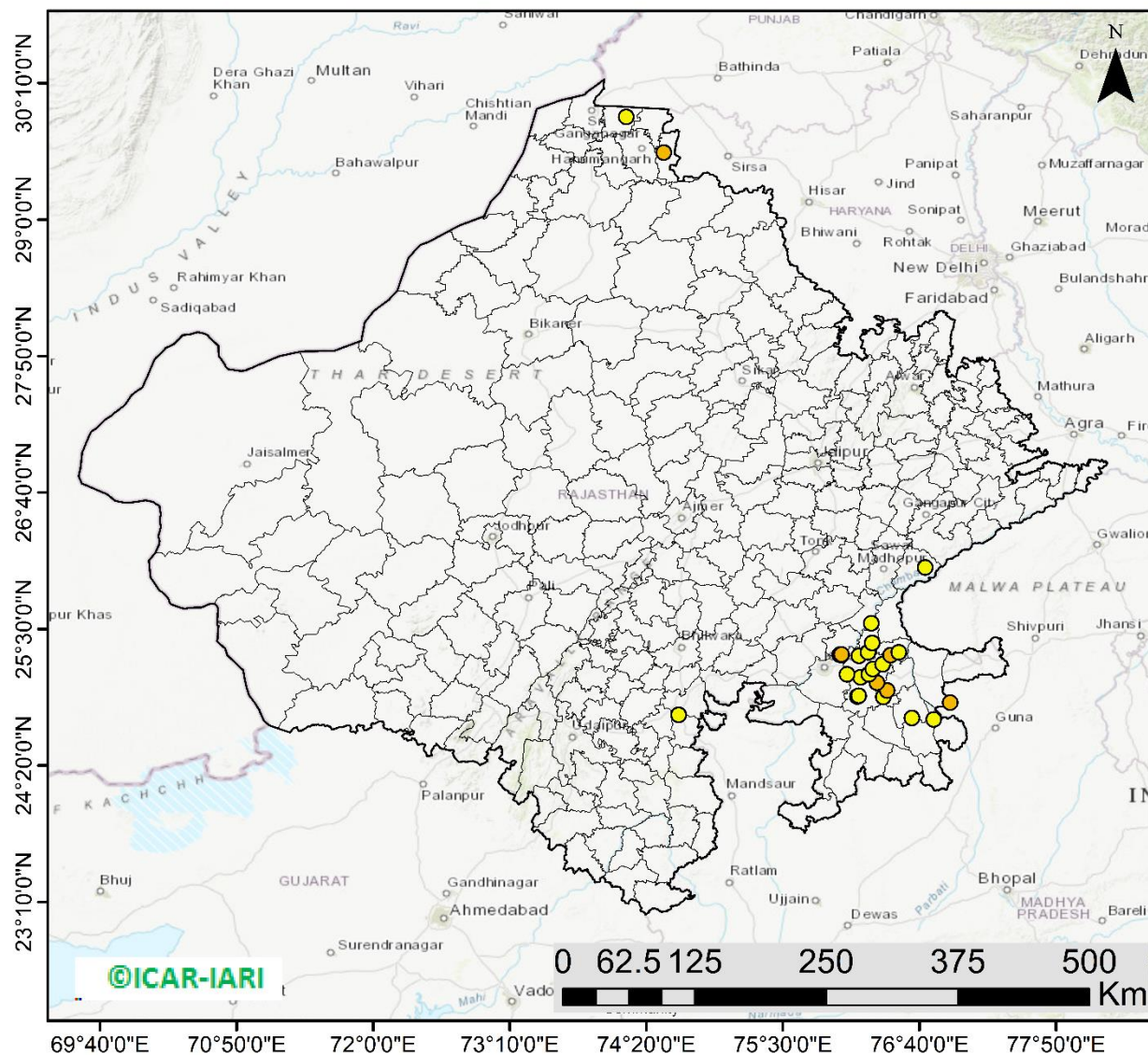
<http://geoportal.icar.gov.in:8080/geoexplorer/composer/>

District-wise cumulative number of residues burning events in Uttar Pradesh (15Sep-10Oct -2023)



Rajasthan

RICE RESIDUE BURNING IN RAJASTHAN



26 burning events
detected in Rajasthan on
(10th Oct 2023)

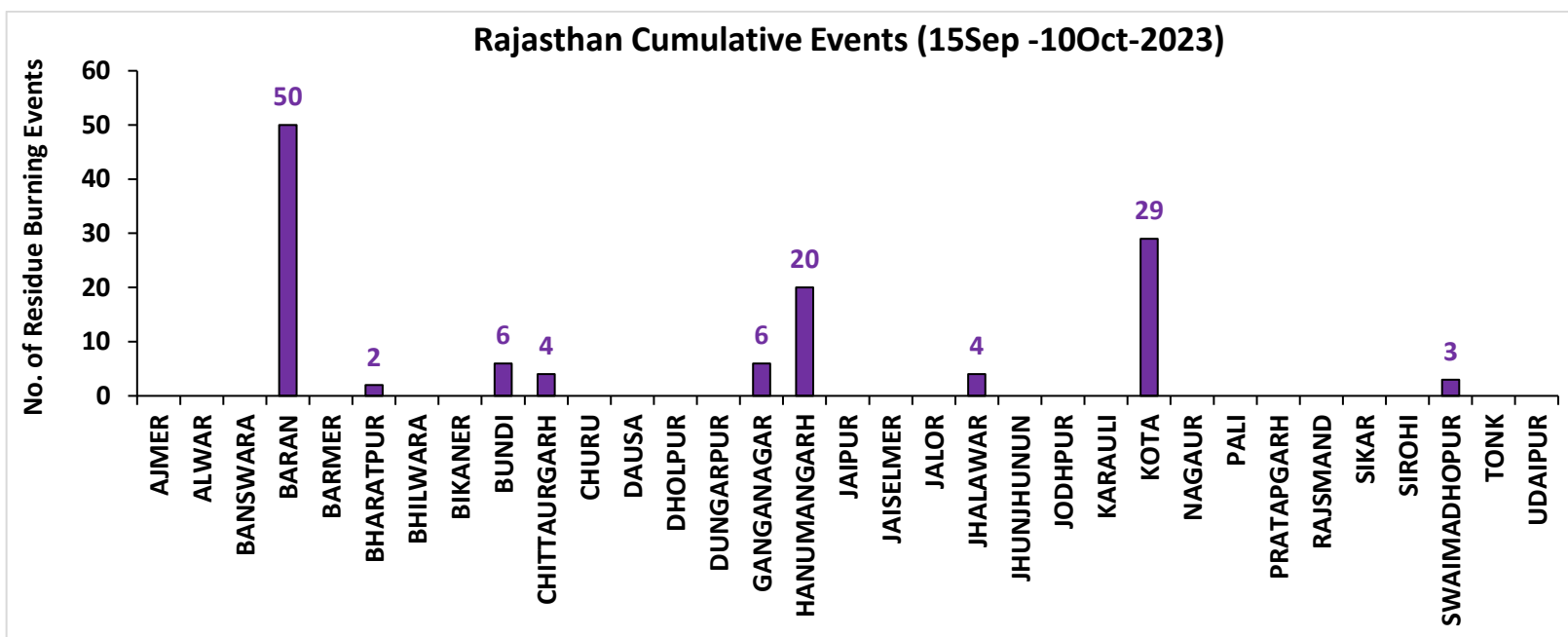
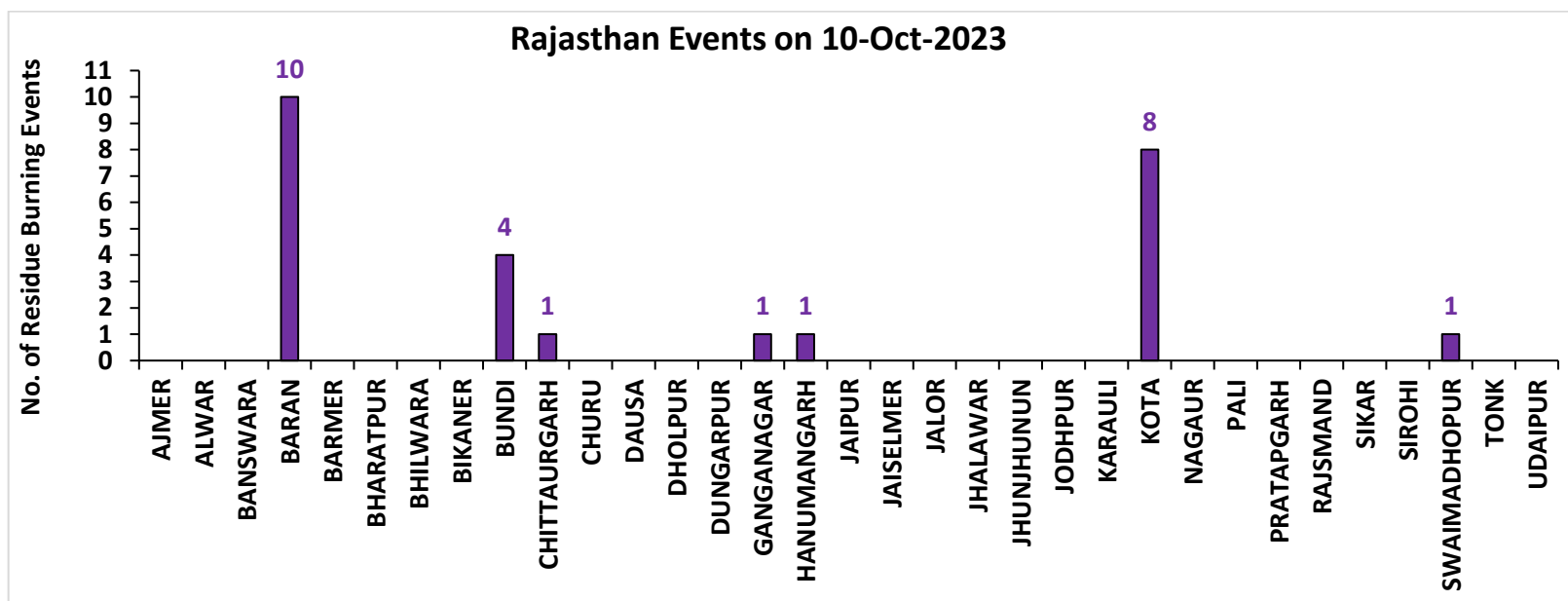
Fire Intensity (W/m²)

- 0 - 5
- 6 - 10
- 11 - 15
- 16 - 20
- >20



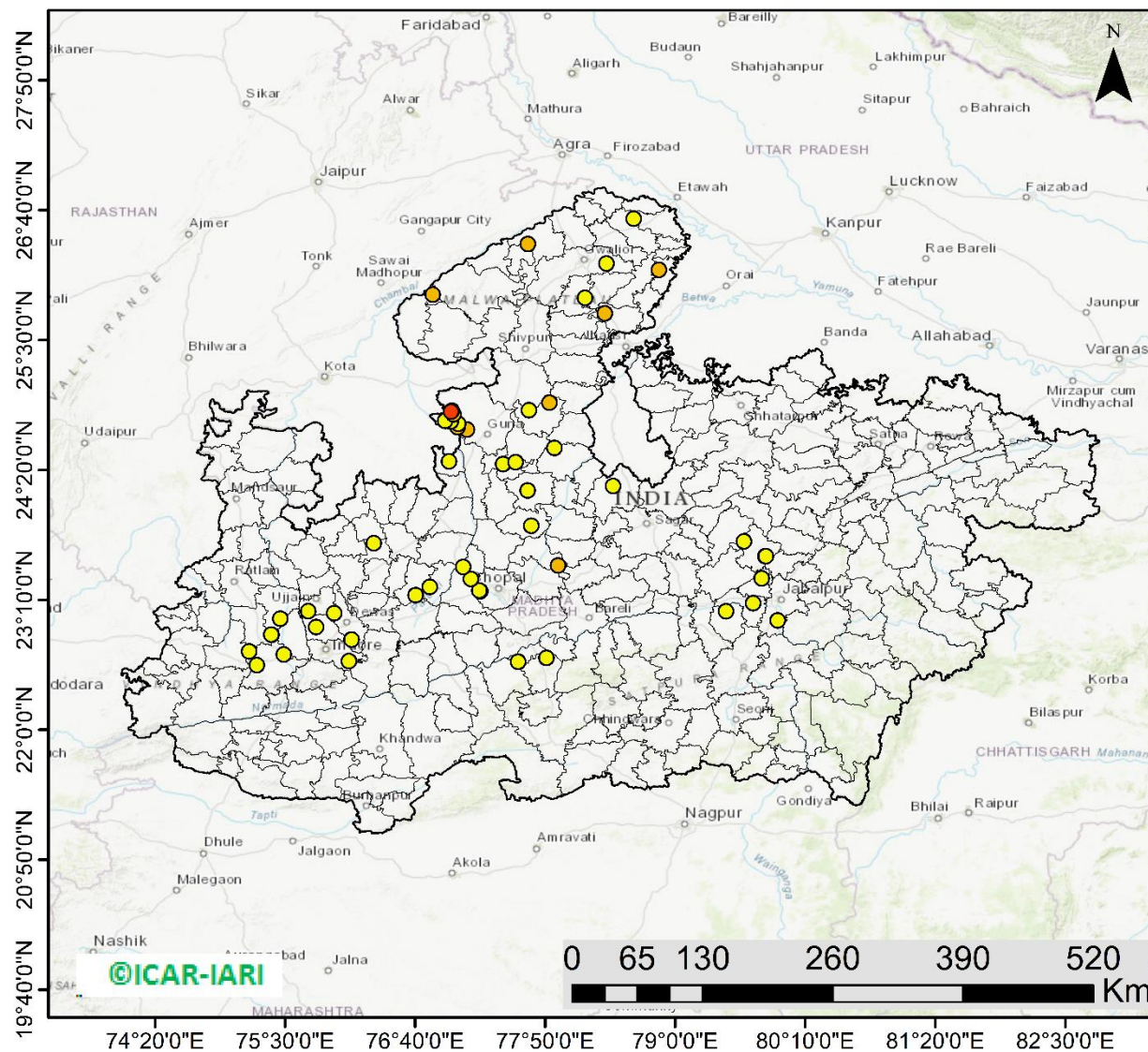
<http://geoportal.icar.gov.in:8080/geoexplorer/composer/>

District-wise cumulative number of residues burning events in Rajasthan (15Sep-10Oct -2023)



Madhya Pradesh

RICE RESIDUE BURNING IN MADHYA PRADESH



52 burning events
detected in MP on
(10th Oct 2023)

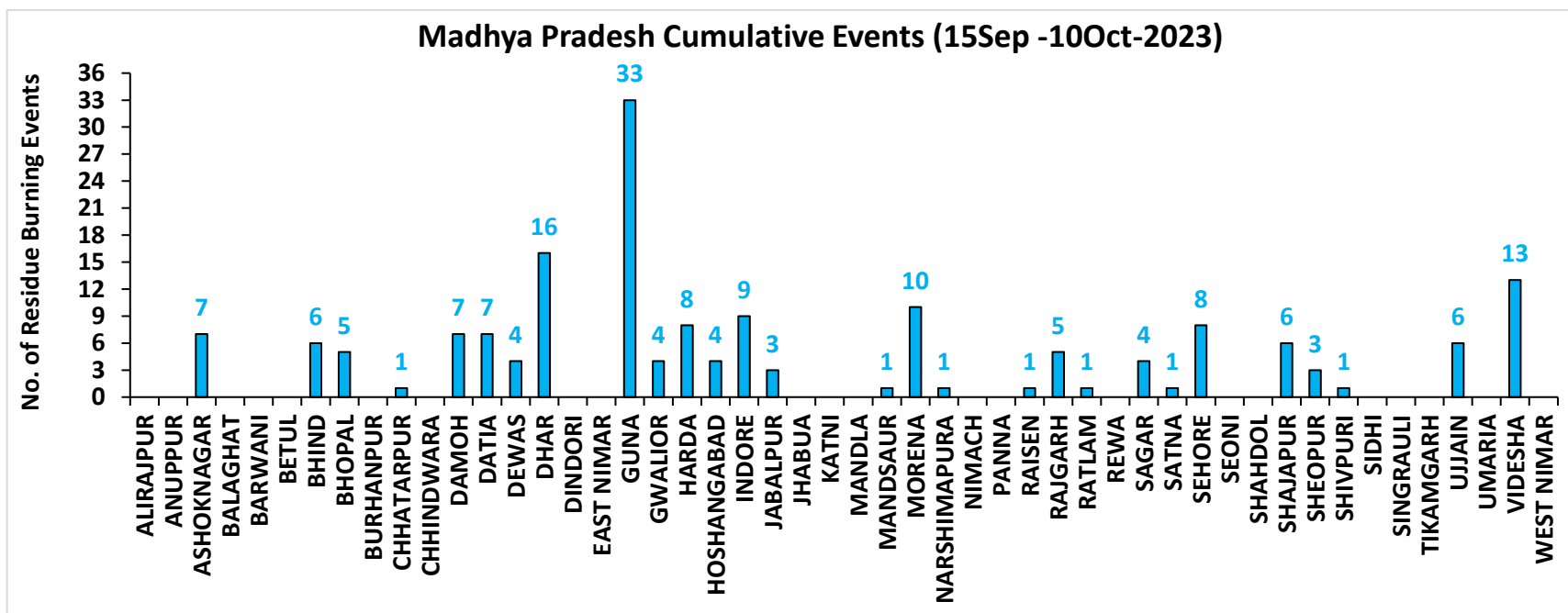
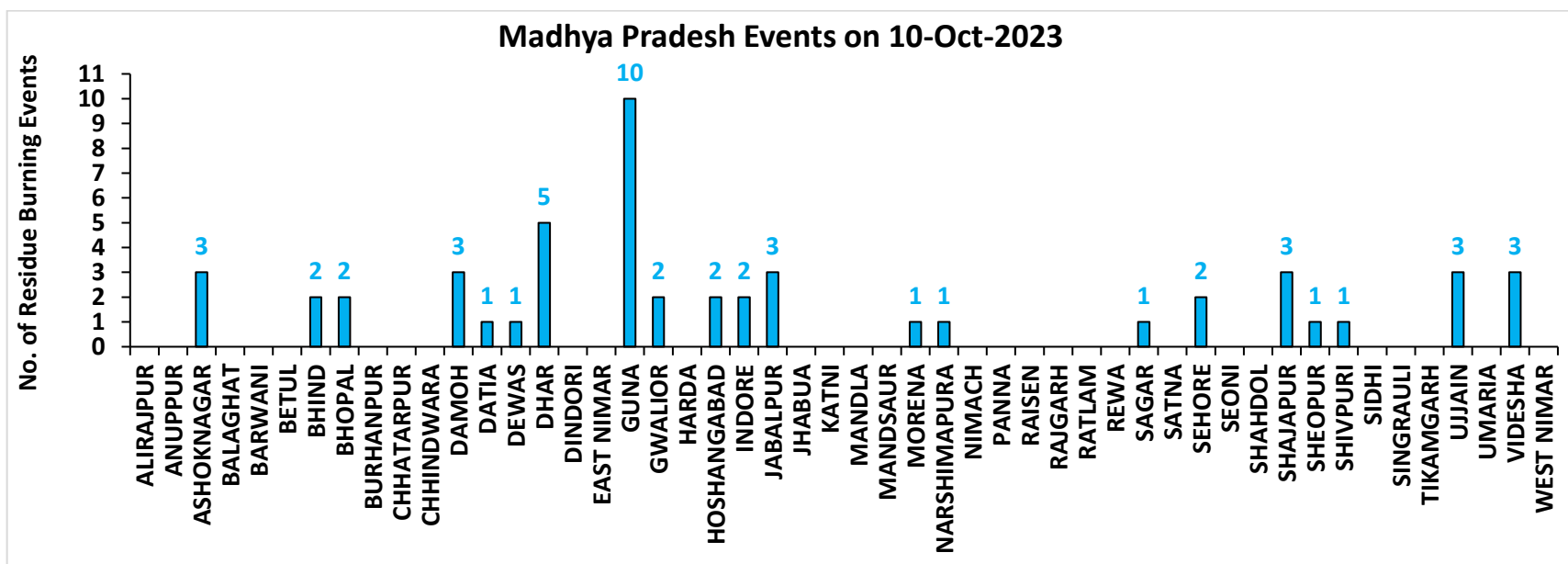
Fire Intensity (W/m²)

- 0 - 5
- 6 - 10
- 11 - 15
- 16 - 20
- >20



<http://geoportal.icar.gov.in:8080/geoexplorer/composer/>

District-wise cumulative number of residues burning events in Madhya Pradesh (15Sep-10Oct -2023)



District-wise cumulative number of residues burning events detected in 2020, 2021, 2022 and 2023
(Period: 15 September-10 October)

(a) Punjab

District	15 Sep-10 Oct			
	2020	2021	2022	2023
Amritsar	1338	318	456	568
Barnala	5	4	3	6
Bathinda	7	0	0	0
Faridkot	22	20	3	8
Fatehgarh Sahib	35	28	4	11
Fazilka	2	3	0	3
Firozpur	76	25	16	25
Gurdaspur	118	20	9	10
Hoshiarpur	5	0	0	2
Jalandhar	30	16	13	16
Kapurthala	44	26	23	61
Ludhiana	67	57	12	9
Mansa	6	1	1	6
Moga	19	7	2	2
Muktsar	2	2	1	0
Pathankot	0	0	0	0
Patiala	175	62	26	88
Rupnagar	10	2	0	1
Sangrur	58	9	12	34
SAS NAGAR (Mohali)	14	11	4	30
SBS Nagar	6	2	2	2
Tarn Taran	541	151	131	155
Total	2580	764	718	1037

(b) Haryana

District	15 Sep-10 Oct			
	2020	2021	2022	2023
AMBALA	48	13	10	59
BHIWANI	0	0	0	1
CHARKHI DADRI	0	0	0	0
FARIDABAD	0	1	0	1
FATEHABAD	22	5	3	40
GURGAON	0	0	0	0
HISAR	4	0	0	13
JHAJJAR	1	0	0	2
JIND	3	6	8	31
KAITHAL	45	43	9	24
KARNAL	83	69	20	20
KURUKSHETRA	71	42	27	49
MAHENDRAGARH	0	0	0	0
MEWAT	0	0	0	0
PALWAL	5	5	0	7
PANCHKULA	0	0	0	0
PANIPAT	6	2	1	11
REWARI	0	0	0	0
ROHTAK	0	0	0	1
SIRSA	2	0	0	1
SONIPAT	6	1	2	31
YAMUNANAGAR	25	9	3	28
Total	321	196	83	319

(c) Uttar Pradesh

District	15 Sep-10 Oct													
	2020	2021	2022	2023	GAUTAMBUDHNAGAR	1	1	0	4	MUZAFARNAGAR	2	0	0	0
AGRA	0	0	0	0	GAZIPUR	0	0	0	0	PILHIBHIT	4	4	2	20
ALIGARH	16	2	1	47	GHAZIABAD	1	2	1	5	PRATAPGARH	0	0	0	0
ALLAHABAD	0	0	0	0	GONDA	0	0	0	0	RAIBEARELI	1	0	0	0
AMBEDKARNAGAR	0	0	0	0	GORAKHPUR	0	0	0	0	RAMPUR	5	14	0	5
AMETHI	0	0	0	0	HAMIRPUR	0	0	0	0	SAMBHAL	6	1	4	8
AMROHA	1	0	0	2	HAPUR	3	0	0	0	SANTKABIRNAGAR	0	0	0	0
AURAIYA	12	0	4	3	HARDOI	2	3	0	3	SANTRAVIDASNAGAR	0	0	0	0
AZAMGARH	0	0	0	0	HATHRAS	0	0	0	2	SHAHJAHANPUR	12	83	53	19
BADAUN	0	0	0	0	JALAUN	0	0	0	0	SHAMLI	17	5	2	2
BAGHPAT	1	0	0	0	JAUNPUR	0	0	0	0	SHARANPUR	14	0	0	0
BAHRAICH	2	0	0	0	JHANSI	5	0	0	2	SHRAWASTI	0	0	0	0
BALLIA	0	0	0	0	KANNAUJ	2	0	0	0	SIDDHARTHANAGAR	0	0	0	0
BALRAMPUR	0	0	0	0	KANPUR	0	1	0	0	SITAPUR	4	1	0	1
BANDA	0	0	0	0	KANPUR DEHAT	1	0	0	0	SONBHADRA	0	0	0	0
BARABANKI	7	0	0	8	KASGANJ	1	0	0	0	SULTANPUR	0	0	0	0
BAREILLY	5	4	0	6	KAUSHAMBI	6	1	0	1	UNNAO	2	0	0	2
BASTI	2	0	0	0	KHERI	15	14	8	1	VARANASI	0	0	0	0
BIJNOR	3	1	1	1	KUSHINAGAR	0	0	0	0	Total	191	150	80	185
BULANDSAHAR	3	1	2	3	LALITPUR	0	1	2	0					
CHANDAULI	0	0	0	0	LUCKNOW	2	0	0	1					
CHITRAKOOT	0	0	0	0	MAHARAJGANJ	0	0	0	0					
DEORIA	0	0	0	2	MAHOBAB	0	0	0	0					
ETAH	4	1	0	7	MAINPURI	0	2	0	3					
ETAWAH	2	3	0	5	MATHURA	4	4	0	16					
FAIZABAD	0	0	0	0	MAU	0	0	0	0					
FARRUKHABAD	0	0	0	0	MEERUT	9	0	0	2					
FATEHPUR	7	0	0	2	MIRZAPUR	5	0	0	0					
FIROZABAD	1	0	0	0	MORADABAD	1	1	0	2					

(d) Delhi

District	15 Sep-10 Oct			
	2020	2021	2022	2023
CENTRAL	0	0	0	0
EAST	0	0	0	0
NEW DELHI	0	0	0	0
NORTH	2	0	2	1
NORTH EAST	0	0	0	0
NORTH WEST DELHI	0	0	0	0
SHAHADRA	0	0	0	0
SOUTH	1	0	0	0
SOUTH EAST	0	0	0	0
SOUTH WEST DELHI	0	0	0	0
WEST DELHI	0	0	0	0
Total	3	0	2	1

(e) Rajasthan

District	15 Sep-10 Oct			
	2020	2021	2022	2023
AJMER	0	0	0	0
ALWAR	1	0	0	0
BANSWARA	0	0	0	0
BARAN	59	0	3	50
BARMER	2	0	0	0
BHARATPUR	0	0	0	2
BHILWARA	2	0	0	0
BIKANER	0	0	0	0
BUNDI	1	0	3	6
CHITTAURGARH	0	1	0	4
CHURU	0	0	0	0
DAUSA	0	0	0	0
DHOLPUR	0	0	0	0
DUNGARPUR	0	0	0	0
GANGANAGAR	2	0	0	6
HANUMANGARH	32	2	2	20
JAIPUR	0	0	0	0
JAISELMER	2	0	0	0
JALOR	0	0	0	0
JHALAWAR	0	0	0	4
JHUNJHUNUN	1	0	0	0
JODHPUR	1	0	0	0
KARAULI	1	0	0	0
KOTA	17	0	1	29
NAGAU	0	0	0	0
PALI	1	0	0	0
PRATAPGARH	0	0	0	0
RAJSMAND	0	0	0	0
SIKAR	0	0	0	0
SIROHI	1	0	0	0
SWAIMADHOPUR	1	0	0	3
TONK	1	0	0	0
UDAIPUR	0	0	0	0
Total	125	3	9	124

(f) Madhya Pradesh

District	15 Sep-10 Oct								
	2020	2021	2022	2023	PANNA	1	0	0	0
ALIRAJPUR	0	0	0	0	RAISEN	2	0	0	1
ANUPPUR	0	0	0	0	RAJGARH	6	1	2	5
ASHOKNAGAR	7	0	0	7	RATLAM	0	0	0	1
BALAGHAT	0	0	0	0	REWA	2	0	0	0
BARWANI	0	0	0	0	SAGAR	9	2	1	4
BETUL	0	0	0	0	SATNA	3	0	0	1
BHIND	0	0	0	6	SEHORE	4	0	1	8
BHOPAL	2	2	0	5	SEONI	2	1	0	0
BURHANPUR	0	0	0	0	SHAHDOL	0	0	0	0
CHHATARPUR	2	0	0	1	SHAJAPUR	4	0	0	6
CHHINDWARA	0	0	0	0	SHEOPUR	3	0	0	3
DAMOH	6	1	0	7	SHIVPURI	3	0	0	1
DATIA	8	2	0	7	SIDHI	0	0	0	0
DEWAS	3	0	0	4	SINGRAULI	0	0	0	0
DHAR	15	0	0	16	TIKAMGARH	1	0	0	0
DINDORI	0	0	0	0	UJJAIN	5	0	0	6
EAST NIMAR	0	1	0	0	UMARIA	0	0	0	0
GUNA	8	1	1	33	VIDESHA	37	5	1	13
GWALIOR	17	0	1	4	WEST NIMAR	1	0	0	0
HARDA	8	7	0	8	Total	178	27	7	175
HOSHANGABAD	5	0	0	4					
INDORE	2	0	0	9					
JABALPUR	6	1	0	3					
JHABUA	0	0	0	0					
KATNI	0	0	0	0					
MANDLA	0	0	0	0					
MANDSAUR	0	0	0	1					
MORENA	5	2	0	10					
NARSHIMAPURA	1	1	0	1					
NIMACH	0	0	0	0					

State-wise and date-wise number of paddy residue burning events detected in 2020, 2021, 2022 and 2023
(Period: 15 September-10 October)

Number of Paddy Residue Burning Events Detected

Punjab					Haryana				Uttar Pradesh			
Date	2020	2021	2022	2023	2020	2021	2022	2023	2020	2021	2022	2023
15-Sep	5	0	14	0	2	0	0	1	1	0	0	0
16-Sep	0	2	0	6	1	0	0	1	0	0	0	2
17-Sep	5	0	1	0	3	0	0	0	0	0	0	0
18-Sep	17	9	7	0	0	0	1	0	0	1	0	0
19-Sep	30	5	18	0	2	0	0	1	0	1	0	0
20-Sep	67	8	10	0	6	0	0	2	2	0	0	0
21-Sep	54	0	56	1	2	0	0	2	1	0	0	1
22-Sep	34	0	30	0	1	0	0	2	1	0	0	0
23-Sep	49	0	3	1	0	0	0	3	1	0	0	0
24-Sep	78	5	0	0	0	0	0	0	1	1	0	1
25-Sep	184	11	0	0	11	0	0	1	5	0	0	1
26-Sep	190	19	0	3	5	0	0	2	8	0	0	0
27-Sep	78	37	0	21	8	0	0	9	4	1	0	0
28-Sep	55	28	0	33	6	1	0	8	1	6	0	4
29-Sep	63	64	0	68	9	4	0	17	6	10	0	6
30-Sep	116	27	8	81	22	2	0	26	13	13	10	18
01-Oct	165	13	45	123	15	4	1	21	7	13	3	17
02-Oct	141	0	83	119	30	2	7	24	12	0	36	7
03-Oct	110	9	75	105	8	6	8	17	4	3	29	8
04-Oct	162	23	65	95	29	2	7	29	8	0	2	9
05-Oct	161	18	130	98	17	3	24	24	14	2	0	14
06-Oct	171	42	85	91	30	0	26	19	12	1	0	18
07-Oct	197	72	62	32	32	11	6	13	41	13	0	5
08-Oct	95	108	19	92	29	27	0	55	25	11	0	27
09-Oct	121	114	3	58	20	39	1	14	15	20	0	13
10-Oct	232	150	4	10	33	95	2	28	9	54	0	34
Total	2580	764	718	1037	321	196	83	319	191	150	80	185

Number of Paddy Residue Burning Events Detected

Delhi					Rajasthan				Madhya Pradesh			
Date	2020	2021	2022	2023	2020	2021	2022	2023	2020	2021	2022	2023
15-Sep	0	0	0	0	1	0	0	0	2	0	0	0
16-Sep	0	0	0	0	0	0	0	0	0	0	0	0
17-Sep	0	0	0	0	0	0	0	0	0	0	0	0
18-Sep	0	0	0	0	0	0	0	0	0	0	0	0
19-Sep	0	0	0	0	0	0	0	0	0	0	0	1
20-Sep	0	0	0	0	0	0	0	0	0	0	0	1
21-Sep	0	0	0	0	1	0	1	0	2	0	0	0
22-Sep	0	0	0	0	0	0	0	0	0	0	0	0
23-Sep	0	0	0	0	1	0	0	0	0	0	0	0
24-Sep	0	0	0	0	0	0	0	0	0	0	0	0
25-Sep	0	0	0	0	1	0	0	0	0	0	0	1
26-Sep	0	0	0	1	1	0	0	0	2	0	0	0
27-Sep	0	0	0	0	0	0	0	0	1	0	0	0
28-Sep	0	0	0	0	0	0	0	0	1	1	0	2
29-Sep	0	0	0	0	3	0	0	6	0	4	0	5
30-Sep	1	0	0	0	1	0	0	4	5	0	0	11
01-Oct	0	0	0	0	4	1	1	4	13	0	1	8
02-Oct	1	0	0	0	7	0	3	4	12	0	1	1
03-Oct	0	0	0	0	9	0	2	5	5	1	5	5
04-Oct	0	0	0	0	15	0	0	22	4	2	0	12
05-Oct	0	0	2	0	14	0	1	11	13	1	0	18
06-Oct	1	0	0	0	16	0	0	9	37	1	0	7
07-Oct	0	0	0	0	14	1	1	7	31	1	0	9
08-Oct	0	0	0	0	11	0	0	9	17	2	0	18
09-Oct	0	0	0	0	12	0	0	17	15	4	0	24
10-Oct	0	0	0	0	14	1	0	26	18	10	0	52
Total	3	0	2	1	125	3	9	124	178	27	7	175

Details of residue burning events in Punjab on 10-Oct-2023

S. No.	District	Tehsil / Block	Satellite	Latitude	Longitude	ACQ_TIME	Day / Night	Fire Power (W/m2)
1	AMRITSAR	AJNALA	S-NPP	31.96645	74.86218	13:28:38	D	1.90
2	FIROZPUR	FIROZPUR	S-NPP	30.94930	74.52304	13:28:38	D	2.20
3	FIROZPUR	MAMDOT	S-NPP	30.79163	74.35879	13:28:38	D	6.00
4	MANSA	SARDULGARH	S-NPP	29.56600	75.25323	13:28:38	D	2.50
5	PATIALA	SAMANA	S-NPP	30.12173	76.09857	13:28:38	D	2.80
6	SANGRUR	ANDANA	S-NPP	29.74740	75.99109	13:28:38	D	2.40
7	SANGRUR	ANDANA	S-NPP	29.79819	76.06277	13:28:38	D	14.00
8	SANGRUR	ANDANA	S-NPP	29.80672	75.98361	13:28:38	D	5.60
9	SANGRUR	ANDANA	S-NPP	29.80877	76.01991	13:28:38	D	2.80
10	TARN TARAN	VALTOHA	S-NPP	31.26111	74.55833	13:28:38	D	4.10

Details of residue burning events in Haryana on 10-Oct-2023

S. No.	District	Tehsil / Block	Satellite	Latitude	Longitude	ACQ_TIME	Day / Night	Fire Power (W/m2)
1	AMBALA	NARAINGARH	S-NPP	30.47129	77.09512	13:28:38	D	3.00
2	BHIWANI	BAWANI KHERA	S-NPP	29.01967	76.08359	13:28:38	D	1.10
3	FARIDABAD	FARIDABAD	S-NPP	28.45351	77.38996	13:28:38	D	3.60
4	FATEHABAD	BHUNA	S-NPP	29.42931	75.70420	13:28:38	D	4.00
5	FATEHABAD	BHUNA	S-NPP	29.60607	75.81939	13:28:38	D	2.70
6	FATEHABAD	FATEHABAD	S-NPP	29.58801	75.41615	13:28:38	D	3.00
7	FATEHABAD	FATEHABAD	AQUA	29.47500	75.53000	13:39:04	D	11.80
8	FATEHABAD	TOHANA	S-NPP	29.63613	75.86253	13:28:38	D	4.60
9	FATEHABAD	TOHANA	S-NPP	29.66853	75.89677	13:28:38	D	4.10
10	FATEHABAD	TOHANA	S-NPP	29.74318	75.87806	13:28:38	D	5.60
11	HISAR	BARWALA	S-NPP	29.35832	75.97303	13:28:38	D	3.20
12	HISAR	BARWALA	S-NPP	29.47765	75.95451	13:28:38	D	1.40
13	HISAR	NARNAUND	S-NPP	29.39209	76.08405	13:28:38	D	1.40
14	HISAR	UKLANA	S-NPP	29.47554	75.96306	13:28:38	D	3.80
15	JIND	ALEWA	S-NPP	29.46778	76.31707	13:28:38	D	3.10
16	JIND	JIND	S-NPP	29.29465	76.45209	13:28:38	D	1.50
17	JIND	NARWANA	S-NPP	29.71442	75.95282	13:28:38	D	5.80
18	JIND	UCHANA	S-NPP	29.41522	76.30694	13:28:38	D	5.70
19	JIND	UCHANA	S-NPP	29.56305	76.31996	13:28:38	D	2.20
20	JIND	UCHANA	S-NPP	29.56911	76.11542	13:28:38	D	8.40
21	KAITHAL	KAITHAL	S-NPP	29.92567	76.46318	13:28:38	D	2.30
22	KAITHAL	KALAYAT	S-NPP	29.55716	76.28036	13:28:38	D	1.40
23	KAITHAL	KALAYAT	S-NPP	29.70276	76.35008	13:28:38	D	2.60
24	KAITHAL	PUNDRI	AQUA	29.63900	76.61300	13:39:04	D	10.90
25	KAITHAL	RAJOUND	S-NPP	29.57069	76.37148	13:28:38	D	1.70
26	KARNAL	KARNAL	S-NPP	29.66885	77.09344	13:28:38	D	1.60
27	PANIPAT	PANIPAT	S-NPP	29.43974	76.93188	13:28:38	D	1.90
28	SONIPAT	MURTHAL	S-NPP	29.06299	77.14049	13:28:38	D	2.90

Details of residue burning events in Uttar Pradesh on 10-Oct-2023

S. No.	District	Tehsil / Block	Satellite	Latitude	Longitude	ACQ_TIME	Day / Night	Fire Power (W/m2)
1	ALIGARH	GABHANA	S-NPP	28.09292	78.02494	13:28:38	D	4.70
2	ALIGARH	KOL	S-NPP	27.73849	78.33347	13:28:38	D	2.50
3	BARABANKI	FATEHPUR	S-NPP	27.18260	81.24979	13:28:38	D	4.30
4	BARABANKI	FATEHPUR	S-NPP	27.24187	81.21119	13:28:38	D	5.90
5	BARABANKI	FATEHPUR	AQUA	27.15300	81.24400	13:39:04	D	5.50
6	BARABANKI	HAIDARGARH	AQUA	26.78100	81.40900	13:39:04	D	9.80
7	BARABANKI	NAWABGANJ	S-NPP	26.85485	81.36497	13:28:38	D	2.70
8	BARABANKI	RAMNAGAR	S-NPP	27.08241	81.31045	13:28:38	D	6.10
9	BAREILLY	BAHERI	S-NPP	28.78368	79.66164	13:28:38	D	7.60
10	BAREILLY	BAHERI	S-NPP	28.78730	79.66106	13:28:38	D	7.80
11	BAREILLY	NAWABGANJ	S-NPP	28.53531	79.65946	13:28:38	D	2.70
12	BIJNOR	CHANDPUR	AQUA	29.05400	78.37400	13:39:04	D	7.70
13	BULANDSAHAR	DIBAI	S-NPP	28.32334	78.15459	13:28:38	D	2.80
14	ETAH	JALESAR	S-NPP	27.51217	78.43040	13:28:38	D	3.60
15	ETAH	JALESAR	S-NPP	27.51443	78.42622	13:28:38	D	2.80
16	ETAH	JALESAR	S-NPP	27.51499	78.43037	13:28:38	D	2.80
17	ETAH	JALESAR	S-NPP	27.51501	78.42568	13:28:38	D	3.60
18	ETAH	JALESAR	S-NPP	27.51559	78.42982	13:28:38	D	3.60
19	ETAH	JALESAR	S-NPP	27.52354	78.41618	13:28:38	D	5.80
20	ETAH	JALESAR	AQUA	27.51800	78.41200	13:39:04	D	11.10
21	ETAWAH	SAIFAI	S-NPP	26.86620	79.00594	13:28:38	D	1.10
22	FATEHPUR	FATEHPUR	S-NPP	25.98512	80.77127	13:28:38	D	3.40
23	FATEHPUR	KHAGA	S-NPP	25.62140	81.17131	13:28:38	D	1.70
24	GAUTAMBUDHNAGAR	JEVAR	S-NPP	28.20210	77.56158	13:28:38	D	6.80
25	HARDOI	BILGRAM	S-NPP	27.11279	80.25241	13:28:38	D	3.30
26	HARDOI	HARDOI	S-NPP	27.38101	79.97491	13:28:38	D	3.40
27	JHANSI	TAHRAULI	S-NPP	25.54826	78.75336	13:28:38	D	3.50
28	MAINPURI	BHOGAON	S-NPP	27.17969	79.15302	13:28:38	D	6.00
29	MAINPURI	BHOGAON	S-NPP	27.18029	79.15376	13:28:38	D	6.10
30	MATHURA	CHHATA	S-NPP	27.63931	77.47733	13:28:38	D	7.50
31	PILHIBHIT	PILHIBHIT	S-NPP	28.81369	79.74774	02:19:21	N	0.60
32	RAMPUR	MILAK	S-NPP	28.62342	79.15710	13:28:38	D	3.00
33	SHAHJAHANPUR	PUWANYA	S-NPP	28.24358	80.17501	02:19:21	N	0.80
34	UNNAO	BIGHAPUR	S-NPP	26.13608	80.68307	13:28:38	D	4.10

Details of residue burning events in Rajasthan on 10-Oct-2023

S. No.	District	Tehsil / Block	Satellite	Latitude	Longitude	ACQ_TIME	Day / Night	Fire Power (W/m2)
1	BARAN	ANTA	S-NPP	25.04020	76.30421	13:28:38	D	7.10
2	BARAN	ANTA	S-NPP	25.11139	76.23556	13:28:38	D	3.50
3	BARAN	ANTA	S-NPP	25.15713	76.27033	13:28:38	D	2.40
4	BARAN	ANTA	S-NPP	25.19744	76.35325	13:28:38	D	1.60
5	BARAN	ATRU	S-NPP	24.73865	76.60361	13:28:38	D	2.10
6	BARAN	BARAN	S-NPP	24.97306	76.39170	13:28:38	D	5.60
7	BARAN	CHHABRA	S-NPP	24.72650	76.78915	13:28:38	D	3.80
8	BARAN	KISHANGUNJ	S-NPP	24.87034	76.93052	13:28:38	D	9.20
9	BARAN	MANGROL	S-NPP	25.27540	76.42154	13:28:38	D	5.40
10	BARAN	MANGROL	S-NPP	25.29993	76.49149	13:28:38	D	3.20
11	BUNDI	INDRAGARH	S-NPP	25.54819	76.25575	13:28:38	D	2.90
12	BUNDI	KESHORAIPATAN	S-NPP	25.27782	76.00259	13:28:38	D	7.90
13	BUNDI	KESHORAIPATAN	S-NPP	25.27871	75.98653	13:28:38	D	9.10
14	BUNDI	KESHORAIPATAN	S-NPP	25.28115	76.00195	13:28:38	D	7.90
15	CHITTAURGARH	CHITTAURGARH	S-NPP	24.76510	74.60995	02:19:21	N	2.10
16	GANGANAGAR	SADULSHAR	S-NPP	29.87686	74.16230	13:28:38	D	0.80
17	HANUMANGARH	TIBI	S-NPP	29.57190	74.48303	13:28:38	D	5.50
18	KOTA	DIGOD	S-NPP	25.08666	76.16525	13:28:38	D	4.10
19	KOTA	DIGOD	S-NPP	25.27042	76.15065	13:28:38	D	2.30
20	KOTA	DIGOD	S-NPP	25.29968	76.22803	13:28:38	D	2.20
21	KOTA	DIGOD	S-NPP	25.37983	76.26415	13:28:38	D	2.90
22	KOTA	LADPURA	S-NPP	25.11193	76.04971	13:28:38	D	5.00
23	KOTA	SANGOD	S-NPP	24.91734	76.35897	13:28:38	D	1.90
24	KOTA	SANGOD	S-NPP	24.92269	76.13726	13:28:38	D	2.70
25	KOTA	SANGOD	S-NPP	24.92784	76.14815	13:28:38	D	2.10
26	SWAIMADHOPUR	KHANDAR	S-NPP	26.02552	76.71706	13:28:38	D	4.20

Details of residue burning events in Madhya Pradesh on 10-Oct-2023

S. No.	District	Tehsil / Block	Satellite	Latitude	Longitude	ACQ_TIME	Day / Night	Fire Power (W/m2)
1	ASHOKNAGAR	ASHOKNAGAR	S-NPP	24.40161	77.56976	13:28:38	D	2.50
2	ASHOKNAGAR	ISAGARH	S-NPP	24.93709	77.87330	13:28:38	D	5.20
3	ASHOKNAGAR	MUNGAOLI	S-NPP	24.53075	77.91747	13:28:38	D	3.00
4	BHIND	BHIND	S-NPP	26.58913	78.62925	13:28:38	D	3.60
5	BHIND	LAHAR	S-NPP	26.12952	78.85485	13:28:38	D	5.30
6	BHOPAL	HUZUR	S-NPP	23.24391	77.25293	13:28:38	D	2.70
7	BHOPAL	HUZUR	S-NPP	23.24937	77.24427	13:28:38	D	2.30
8	DAMOH	JABERA	S-NPP	23.55768	79.81058	13:28:38	D	1.40
9	DAMOH	JABERA	S-NPP	23.55825	79.81471	13:28:38	D	3.40
10	DAMOH	JABERA	S-NPP	23.69063	79.61825	13:28:38	D	1.80
11	DATIA	DATIA	S-NPP	25.73813	78.37027	13:28:38	D	5.80
12	DEWAS	DEWAS	S-NPP	22.80891	76.09731	13:28:38	D	2.90
13	DHAR	BADNAWAR	S-NPP	22.85177	75.37945	02:19:21	N	0.60
14	DHAR	BADNAWAR	S-NPP	22.85350	75.37705	02:19:21	N	0.60
15	DHAR	DHAR	S-NPP	22.57901	75.24751	13:28:38	D	2.20
16	DHAR	DHAR	S-NPP	22.67461	75.48911	13:28:38	D	0.70
17	DHAR	DHAR	S-NPP	22.70257	75.17866	13:28:38	D	3.00
18	GUNA	ARON	S-NPP	24.38465	77.45647	13:28:38	D	3.10
19	GUNA	GUNA	S-NPP	24.69483	77.13338	13:28:38	D	7.20
20	GUNA	GUNA	S-NPP	24.71582	77.04796	13:28:38	D	5.80
21	GUNA	GUNA	S-NPP	24.74663	77.05375	13:28:38	D	2.60
22	GUNA	GUNA	S-NPP	24.76716	76.98779	13:28:38	D	3.00
23	GUNA	GUNA	S-NPP	24.77288	76.93627	13:28:38	D	4.70
24	GUNA	GUNA	S-NPP	24.83173	77.01154	13:28:38	D	7.30
25	GUNA	GUNA	S-NPP	24.86363	76.99792	13:28:38	D	9.30
26	GUNA	GUNA	AQUA	24.85600	76.99000	13:39:04	D	15.60
27	GUNA	KUMBHRAJ	S-NPP	24.40948	76.97221	13:28:38	D	3.50
28	GWALIOR	DABRA	S-NPP	25.87736	78.19274	13:28:38	D	3.20
29	GWALIOR	GWALIOR	S-NPP	26.18601	78.38473	13:28:38	D	2.50
30	HOSHANGABAD	HOSHANGABAD	S-NPP	22.60945	77.59286	13:28:38	D	2.20

31	HOSHANGABAD	ITARSI	S-NPP	22.64444	77.84650	13:28:38	D	2.30
32	INDORE	INDORE	S-NPP	22.61596	76.07349	13:28:38	D	2.40
33	INDORE	SANWER	S-NPP	22.92207	75.77927	13:28:38	D	3.40
34	JABALPUR	JABALPUR	S-NPP	22.98046	79.92031	13:28:38	D	3.20
35	JABALPUR	PATAN	S-NPP	23.35878	79.77773	13:28:38	D	3.50
36	JABALPUR	SAHPURA	S-NPP	23.13589	79.70019	13:28:38	D	3.90
37	MORENA	JORA	S-NPP	26.36114	77.67980	13:28:38	D	5.70
38	NARSHIMAPURA	GOTECALON	S-NPP	23.06356	79.45740	13:28:38	D	1.80
39	SAGAR	KHURAI	S-NPP	24.18839	78.44386	13:28:38	D	2.50
40	SEHORE	SEHORE	S-NPP	23.35294	77.16779	13:28:38	D	3.10
41	SEHORE	SEHORE	S-NPP	23.46058	77.09864	13:28:38	D	2.40
42	SHAJAPUR	KALA PIPAL	S-NPP	23.28198	76.79858	13:28:38	D	2.70
43	SHAJAPUR	MOMAN BARODIY	S-NPP	23.67408	76.29664	13:28:38	D	1.80
44	SHAJAPUR	SUJALPUR	S-NPP	23.20820	76.67242	13:28:38	D	1.90
45	SHEOPUR	SHEOPUR	S-NPP	25.90677	76.82562	13:28:38	D	5.10
46	SHIVPURI	KOLARAS	S-NPP	24.86913	77.68945	13:28:38	D	4.20
47	UJJAIN	BARNAGAR	S-NPP	22.99704	75.45735	02:19:21	N	0.40
48	UJJAIN	UJJAIN	S-NPP	23.04646	75.94135	13:28:38	D	2.30
49	UJJAIN	UJJAIN	S-NPP	23.06406	75.70760	13:28:38	D	3.20
50	VIDESHA	NATERAN	S-NPP	23.83110	77.71006	13:28:38	D	2.90
51	VIDESHA	SIRANJ	S-NPP	24.14773	77.67732	13:28:38	D	1.70
52	VIDESHA	VIDESHA	S-NPP	23.47448	77.95013	13:28:38	D	6.10

The study used data received from following Satellites at NRSC Ground Station

S. No.	Satellite Name	Sensor	Resolution (meter)	Day / Night Passes
1.	Suomi NPP	VIIRS	375 / 750	Both
2.	Terra	MODIS	1000	Both
3.	Aqua	MODIS	1000	Both

**The rice mask for the Punjab and Haryana were provided by MNCFC, MoA&FW, Govt of India
The kharif crop mask for Delhi, Uttar Pradesh, Rajasthan and Madhya Pradesh were provided by NRSC (ISRO)**

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GIS Maps of fire events can be visualized online on ICAR KRISHI Geoportal website: <http://geoportal.icar.gov.in:8080/geoexplorer/composer/>
(Part of KRISHI Portal: <https://krishi.icar.gov.in> initiative)
